BearMamba-3: Mamba-3 State-Space Bearing Fault Diagnosis with Kinematic Frequency Inductive Bias and Multi-Sensor Fusion

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Abstract

Reliable diagnostic instrumentation for rolling bearings demands accurate and interpretable inference from vibration measurements. State-space models (SSMs) have recently achieved competitive accuracy on vibration-based bearing fault diagnosis, yet embedding physical domain knowledge at the activation level remains largely unexplored. This paper presents **BearMamba-3**, a bearing diagnostic framework built on the official Mamba-3 backbone (ICLR 2026), augmented with a *kinematic frequency inductive bias* loss $\mathcal{L}_{\text{kin}}$ and a multi-sensor convolutional input pathway.

$\mathcal{L}_{\text{kin}}$ exploits a property unique to Mamba-3's complex oscillatory states: each state dimension carries an instantaneous rotation frequency $f_{h,k}(t)$, interpretable as a physical resonance in Hz. The loss aligns these state frequencies with fault harmonics computed per sample from the shaft speed — a constraint undefined for real-valued Mamba-2, structurally binding the Mamba-3 backbone to the kinematic loss design.

Experiments across CWRU, Paderborn University (PU), and XJTU-SY yield four findings: *(1)* BearMamba-3 achieves accuracy non-inferior to BearMamba-2 ($p > 0.05$, $n = 5$, cross-condition PU), confirming Mamba-3 does not trade accuracy for interpretability; *(2)* $\mathcal{L}_{\text{kin}}$ reduces inter-seed variance by 14–62% under AWGN on CWRU (descriptive; $p \geq 0.25$), and frequency snapshots visualise progressive fault-harmonic alignment during training; *(3)* dual-sensor fusion yields a statistically significant $+0.70$ pp gain at SNR $= -4$ dB on CWRU (Wilcoxon $p = 0.031$, $n = 5$), while cross-condition inner-race recall improves $+31.6$ pp on XJTU-SY (5/5 seeds, $p = 0.063$), characterising a precision–sample-efficiency trade-off; *(4)* a cross-dataset BatchNorm ablation reveals that 1D-CNN's within-condition advantage over BM3 is principally attributable to BatchNorm regularisation (removing BatchNorm reduces 1D-CNN by 12.7 pp at -8 dB, $p = 0.063$; without BatchNorm, $p = 0.125$ vs BM3), whereas BM3 $+\mathcal{L}_{\text{kin}}$ achieves $+38.4$ pp macro-F1 on XJTU-SY cross-condition

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evaluation (5/5 seeds, $p = 0.063$), demonstrating that physical frequency alignment provides out-of-distribution robustness that statistical normalisation cannot replicate. Code and configurations are publicly available.

Index Terms

Bearing fault diagnosis, state-space model, Mamba-3, physical inductive bias, kinematic loss, multi-sensor fusion, instantaneous frequency, vibration measurement, diagnostic instrumentation.

I. Introduction

Rolling element bearings are among the most failure-prone components in rotating machinery; their unexpected breakdown accounts for a disproportionate share of unplanned industrial downtime [1]. Reliable on-line vibration measurement and condition monitoring are therefore central to predictive maintenance instrumentation, and data-driven fault diagnosis—the automatic identification of fault location (inner race, outer race, rolling element, cage) from vibration signals—has attracted sustained research attention over the past decade [2].

a) From CNNs to sequence models.: Early deep-learning methods relied on 1-D and 2-D convolutional neural networks that treat a sliding vibration window as a fixed-length pattern classification problem [3], [4]. Transformers subsequently demonstrated that long-range temporal modelling can capture fault-periodic harmonics more faithfully than local convolutions [5], but their quadratic self-attention cost scales poorly with window length. State-space models (SSMs) offer a compelling middle ground: linear-time sequence processing with a principled recurrent mechanism that can, in principle, encode periodicity through its transition eigenstructure [6]. Our prior work introduced BearMamba-2, an SSM bearing diagnostic model using the Mamba-2 backbone that achieves competitive accuracy under AWGN and cross-condition transfer, and showed that a complex-state extension (KG-cSSM) further benefits variable-speed diagnosis through architecture-level physics priors.

b) Gap: activation-level physical alignment for Mamba-3.: Recent work has embedded physical priors into bearing diagnostics at different levels of the model: Jiang et al. [7] guide feature alignment via mechanism-driven domain adaptation at the distribution level; PINNs [8], [9] impose governing equations on network outputs. However, none of these approaches can leverage the Mamba-3 architecture's unique capability: *per-step oscillatory states* that carry instantaneous physical frequencies as first-class activation values. The Mamba-3 architecture [10] parameterises each state dimension by an instantaneous angle activation $\theta_{h,k}(t)$ projected from the current input, yielding a sequence of instantaneous physical frequencies $f_{h,k}(t)$ (measured in Hz) interpretable as the momentary resonance frequency of each state at each time step. We observe that $f_{h,k}(t)$ provides a direct *activation-level hook* for injecting bearing kinematics: by encouraging the model's internal oscillation frequencies to align with the analytically known fault harmonics (f_{BPFO}, f_{BPFI}, BSF, and their harmonics — derived in Section III), we endow the model with a physically meaningful inductive bias without modifying its architecture or adding parameters at inference time.

c) Gap: multi-sensor Mamba-3.: Practical deployment often provides more than one vibration channel: drive-end and fan-end accelerometers in lab rigs, or orthogonal horizontal and vertical axes in industrial field sensors. Multi-sensor fusion at the convolutional embedding stage is architecturally trivial but its *interaction* with the SSM state bandwidth (`mimo_rank` in Mamba-3) has not been characterised. Understanding when additional sensor channels improve diagnosis — and when they induce negative transfer in low-data regimes — is practically important.

d) Contributions.: This paper proposes **BearMamba-3**, a bearing fault diagnosis framework with the following four contributions:

C1 First Mamba-3 bearing diagnostic model with structural binding to physical frequency alignment. We demonstrate that the ICLR 2026 official Mamba-3 implementation [10] can be effectively applied to bearing diagnosis, and—crucially—that its oscillatory state mechanism is the *unique carrier* of our kinematic loss $\mathcal{L}_{\text{kin}}$: real-valued SSMs (Mamba-2) lack instantaneous rotation frequencies, rendering $\mathcal{L}_{\text{kin}}$ architecturally undefined for them. The adoption of Mamba-3 and the design of $\mathcal{L}_{\text{kin}}$ are therefore a single integrated decision (Section IV-B).

C2 $\mathcal{L}_{\text{kin}}$: activation-level kinematic inductive bias. We formulate a lightweight auxiliary loss that penalises the distance between each model state's instantaneous frequency and the nearest bearing fault harmonic, computed per sample from the actual shaft speed. $\mathcal{L}_{\text{kin}}$ requires no additional parameters, is computed only during training, and naturally accommodates variable-speed signals via per-sample RPM targets. Empirically, $\mathcal{L}_{\text{kin}}$: *(i)* improves physical interpretability by aligning state frequencies with fault harmonics, quantified through I1 snapshots (Section V-D2); *(ii)* reduces inter-seed accuracy variance under AWGN (std $\downarrow$14–62% across the EAAI evaluation SNR grid for CWRU; Section V-D1), a descriptive observation without statistical significance claim at $\alpha = 0.05$; *(iii)* reveals the boundary conditions of activation-level physics priors: ineffective in discrete-speed regimes (Paderborn) and amplitude-dominated run-to-failure signals (XJTU-SY).

C3 Multi-sensor fusion with precision–sample-efficiency trade-off characterisation. We extend BearMamba-3 to multi-sensor input via a shared convolutional embedding (n_s input channels) and ablate the MIMO–sensor-count interaction through systematic experiments on CWRU (drive-end + fan-end) and XJTU-SY (horizontal + vertical). **The sole statistically significant finding** is a dual-sensor accuracy gain of $+0.70$ pp at SNR $= -4$ dB on CWRU (Wilcoxon $p = 0.031$, $n = 5$, one-sided). XJTU-SY experiments provide complementary mechanistic evidence of a precision–sample-efficiency trade-off: dual-sensor cross-condition inner-race recall improves by $+31.6$ pp (5/5 seeds concordant, $p = 0.063$, above $\alpha = 0.05$), while leave-one-bearing-out with a single OR training instance induces negative transfer (-15.5 pp, $p = 0.063$, 5/5 seeds negative). The XJTU results are reported as mechanistic evidence, not as statistically significant claims (Section V-E2).

C4 Cross-paradigm analysis: BN-dependent vs. frequency-grounded inductive bias. A systematic BatchNorm ablation on 1D-CNN reveals that BatchNorm is the principal source of 1D-CNN's within-condition low-SNR advantage (-12.7 pp without BatchNorm at -8 dB, $p = 0.063$; gap with BM3 becomes non-significant, $p = 0.125$). Concurrently, BearMamba-3 $+\mathcal{L}_{\text{kin}}$ achieves $+38.4$ pp macro-F1 on XJTU-SY cross-condition OOD evaluation (5/5 seeds, $p = 0.063$) over 1D-CNN, which collapses to a constant-OR predictor. This exposes a systematic complementarity: the *BN-dependent paradigm* confers powerful within-distribution regularisation at the cost of OOD brittleness; the *frequency-grounded paradigm* provides distribution-agnostic physical constraints at no within-condition cost once BatchNorm is absent. (Section V-F)

e) Relationship to prior work.: BearMamba-3 is positioned in a progression of increasingly deep physics integration: feature-level noise-awareness (BearMamba-2, submitted) $\rightarrow$ architecture-level complex-state priors for variable-speed diagnosis (KG-cSSM, under review) $\rightarrow$ **BearMamba-3** (activation-level kinematic inductive bias $\times$ multi-sensor). BearMamba-3 is differentiated from the KG-cSSM line by using the official Mamba-3 backbone [10] (rather than a custom complex-state design), by per-sample variable-speed loss computation, and by the systematic multi-sensor ablation.

f) Experimental scope.: All experiments use 5 independent random seeds with Wilcoxon signed-rank tests ($n = 5$). On CWRU, we benchmark against a classical baseline (SK-SVM [11]), two deep learning baselines (1D-CNN and Transformer-1D), and BearMamba-2 (Table III); focused ablation studies then isolate the Mamba-3 vs Mamba-2 and $\pm\mathcal{L}_{\text{kin}}$ effects across the EAAI SNR evaluation grid (-8 to $+10$ dB).

g) Paper organisation.: Section II reviews related work. Section III provides the bearing kinematics background and defines the fault characteristic frequency equations used throughout. Section IV describes the BearMamba-3 architecture and $\mathcal{L}_{\text{kin}}$. Section V presents experimental results on CWRU, Paderborn, and XJTU-SY. Section VI discusses findings, boundary conditions, and limitations. Section VII concludes.

II. Related Work

A. Classical Signal Processing Baselines

Classical bearing diagnosis pipelines combine signal processing with shallow classifiers: envelope analysis demodulates the carrier to expose fault-frequency sidebands, spectral kurtosis (SK) identifies the optimal demodulation band, and hand-crafted statistical features (RMS, kurtosis, crest factor) feed linear SVMs or k-nearest-neighbour classifiers [11], [12]. These methods are interpretable and require no GPU, but their feature sets are fixed at design time and do not generalise to variable shaft speed without explicit per-sample recomputation of fault frequencies. We include a representative SK-SVM baseline in our CWRU evaluation (Table III) to provide an absolute accuracy anchor and to highlight where deep models, and BearMamba-3 in particular, add value beyond hand-crafted features.

B. *Deep Learning for Bearing Fault Diagnosis*

Convolutional neural networks (CNNs) established a dominant paradigm for bearing fault diagnosis: 1-D convolutions extract periodic impulse features directly from raw vibration waveforms, while 2-D convolutions operate on time-frequency representations such as spectrograms and scalograms [3], [4], [13], [14]. Variants with dilated kernels, multi-scale receptive fields, and residual connections have progressively improved accuracy under clean-signal conditions and cross-domain scenarios [15], [16], [17], [18].

Attention mechanisms and Transformer encoders were subsequently applied to bearing diagnosis, with self-attention providing a principled mechanism for relating fault-periodic harmonics across long windows [5], [19]. However, the quadratic attention cost limits practical window length, and the position encodings of vanilla Transformers lack explicit knowledge of bearing kinematics. Hybrid CNN-attention architectures and attention-based transfer learning methods have partially addressed these limitations [20], [21]. Comprehensive surveys of deep learning and transfer learning for bearing diagnosis are provided in [22], [23].

A parallel line of work incorporates *physical prior knowledge* at the signal-processing level, using envelope analysis, EEMD, and STFT to pre-extract fault-frequency features before feeding them to classifiers [12], [24]. Our approach instead embeds kinematic knowledge as a soft regulariser within end-to-end training, preserving the flexibility of raw-waveform learning while providing physically motivated frequency alignment.

C. *State-Space Models and Mamba for Fault Diagnosis*

Structured state-space models (SSMs) [6] — including S4, S5, H3, and Mamba — have emerged as competitive alternatives to Transformers for long-sequence modelling, combining linear-time complexity with rich temporal representations. Mamba [25] introduced input-dependent selective state updates, and Mamba-2 [26] further simplified the formulation by relating SSM layers to multi-head linear attention. Mamba-3 [10] generalises Mamba-2 by replacing scalar real states with complex-valued oscillatory states governed by per-step angle activations, enabling explicit frequency-domain modelling in the state update.

The efficiency of Mamba on long sequences [25], [26] has been validated across diverse time-series tasks [27], and the foundational HiPPO memory projection underpinning SSMs has been shown to capture long-range dependencies robustly [28]. Prior bearing SSM work introduced BearMamba-2 (Mamba-2 backbone, feature-level noise-awareness, submitted) and KG-cSSM (complex-state priors at the architecture level for variable-speed diagnosis, under review). Concurrently, Li *et al.* [29] propose PG-TMT, which embeds physics guidance via frequency-domain attention alignment with bearing fault-order bands, achieving sub-1 MB edge deployment and EVT-based reliability calibration; their work positions on operational reliability and deployment efficiency rather than within-condition accuracy maximisation, and avoids direct comparison against classical 1D-CNN baselines. In contrast, the present work introduces physics directly at the SSM state level: the kinematic regulariser $\mathcal{L}_{\text{kin}}$ acts on Mamba-3's complex eigenvalues, which encode oscillation frequency through their phase angle. These represent two complementary physics-guidance strategies—*soft post-hoc alignment* versus *hard architectural embedding*—targeting different deployment regimes (edge-IoT versus industrial PC with cross-condition robustness). The present work is the first to: (i) employ the official Mamba-3 backbone in a bearing diagnostic context; (ii) exploit the instantaneous angle activations of Mamba-3 states as the hook for a per-sample kinematic loss function.

D. Physical Inductive Biases in Neural Diagnostics

Physics-informed neural networks (PINNs) [8], [9] impose governing equations as soft constraints in the training loss, typically acting on the network's outputs. In fault diagnosis, physical constraints have been applied at various levels: physics-inspired correlation fusion integrating domain knowledge into feature alignment [30], noise-robust layerwise feature learning with structured constraints on the loss landscape [31], and domain-adversarial ensemble learning that penalises domain-inconsistent intermediate representations [32].

Most closely related to our work, Jiang et al. [7] proposed a mechanism-driven domain adaptation model in which bearing fault physics are embedded as inductive biases to guide feature alignment across operating conditions, simultaneously improving cross-domain accuracy and interpretability. Our $\mathcal{L}_{\text{kin}}$ shares the spirit of making physical mechanism knowledge a first-class constraint, but differs in two key respects: (i) it operates at the *activation level* of a Mamba-3 state-space model — aligning intermediate *instantaneous oscillatory state frequencies* with analytically known fault harmonics — rather than at the output or domain-alignment level; (ii) it is architecturally bound to the Mamba-3 backbone (Section IV-B), making the backbone selection and physical prior design a single integrated decision rather than two independent components. This activation-level operation distinguishes $\mathcal{L}_{\text{kin}}$ from output-level PINNs [8], [9] (which guide predictions, not intermediate features) and from weight-level or loss-level regularisers (which constrain parameters or domain alignment, not per-state frequency activations). Unlike prior work that attracted learned features toward a single fault class [33], our cover variant generalises to all fault frequency families and to variable-speed per-sample targets.

E. Multi-sensor Fusion in Rotating Machinery Diagnosis

Early sensor fusion for fault diagnosis concatenated multiscale convolutional features from multiple sensor locations before a classifier [34], [35]. More recent approaches fuse at the waveform level through sparse autoencoder and deep belief network architectures [36], [37], attention-based channel alignment [38], [39], or multi-sensor residual convolutional fusion under noisy conditions [40]. Multi-sensor 2-D CNN fusion has demonstrated accuracy improvements when sufficient multi-position data are available [41].

Orthogonal-axis accelerometers (horizontal and vertical) have been shown to provide complementary fault signatures for different fault types [42]: inner-race faults excite strong vibrations in both directions, while outer-race faults may be more pronounced in a single directional axis depending on load zone alignment. This physical asymmetry directly motivates our finding (Section V-E2) that dual-sensor fusion benefits IR recall much more than OR recall in leave-one-bearing-out evaluation.

The precision–sample-efficiency trade-off we characterise — additional sensor channels improve performance when training data is sufficient but induce negative transfer in severe few-shot regimes — has been observed in multi-sensor graph-network studies [43] and adaptive viewable-graph fusion under adverse conditions [44], but has not been systematically documented for SSM-based bearing diagnostics.

Summary and Research Gaps

In summary, the existing literature reveals two open problems addressed by this work. First, while SSMs with complex-valued states carry instantaneous oscillatory frequencies that have natural physical semantics in bearing diagnostics, no prior work has exploited these *activation-level* frequencies as a hook for per-sample kinematic regularisation — existing physics-informed approaches operate at the output, domain-alignment, or weight level. Second, the interaction between SSM state bandwidth and multi-sensor input count — and its dependence on available training data — has not been characterised. BearMamba-3 addresses both gaps through an integrated design in which the Mamba-3 backbone selection and the kinematic loss are structurally bound, and through systematic single- versus dual-sensor ablations across three public datasets.

III. Physical Background

A. Bearing Fault Characteristic Frequencies

Rolling element bearings generate periodic vibration signatures whose frequencies are analytically determined by bearing geometry and shaft speed. For a bearing with n_b rolling elements, ball diameter

d, pitch diameter D, and contact angle α, the four primary fault characteristic frequencies are [12], [24]:

$$f_{\text{BPFO}} = \frac{n_b}{2}\, f_r \left(1 - \frac{d}{D}\cos\alpha\right), \tag{1}$$

$$f_{\text{BPFI}} = \frac{n_b}{2}\, f_r \left(1 + \frac{d}{D}\cos\alpha\right), \tag{2}$$

$$f_{\text{BSF}} = \frac{D}{2d}\, f_r \left[1 - \left(\frac{d}{D}\cos\alpha\right)^2\right], \tag{3}$$

$$f_{\text{FTF}} = \frac{f_r}{2} \left(1 - \frac{d}{D}\cos\alpha\right), \tag{4}$$

where $f_r = \Omega/60$ is the shaft rotation frequency (Hz) and Ω is shaft speed (rpm). f_{BPFO} (Ball Pass Frequency, Outer Race) is excited by outer-race defects; f_{BPFI} by inner-race defects; f_{BSF} (Ball Spin Frequency) by rolling element defects; and f_{FTF} (Fundamental Train Frequency) by cage defects. Each characteristic frequency and its integer harmonics produce periodic impulse trains in the measured vibration signal, which form the primary diagnostic signatures for bearing condition monitoring and fault detection [2], [11].

Table I lists the bearing parameters used in this study and the resulting characteristic frequencies at nominal shaft speeds.

TABLE I

BEARING PARAMETERS AND FAULT CHARACTERISTIC FREQUENCIES AT NOMINAL SHAFT SPEED FOR EACH DATASET.

Dataset	Bearing	n_b	d (mm)	D (mm)	Ω (rpm)	f_{BPFO} / f_{BPFI} (Hz)
CWRU DE	SKF 6205	9	7.94	39.04	1797	107.36 / 162.19
CWRU FE	SKF 6203	8	6.75	28.50	1797	$91.44^\dagger$ / $148.16^\dagger$
PU	SKF 6203	8	6.75	29.05	1500	76.77 / 123.23
XJTU-SY	LDK UER204	8	7.92	34.55	2400	123.32 / 196.68

CWRU FE bearing (SKF 6203) has a different pitch diameter (28.50 mm) from the PU bearing of the same part number (29.05 mm); these are computed separately. All contact angles $\alpha = 0°$.

B. Speed-Dependent Shift and Per-Sample Frequency Targets

Because all characteristic frequencies scale proportionally with shaft speed f_r, any variation in operating speed directly shifts the entire fault frequency spectrum. Under variable-speed conditions — such as the accelerated-life XJTU-SY recordings where instantaneous speed fluctuates within a measurement window — using a fixed dataset-level frequency set would introduce misalignment errors proportional to the speed deviation [45], [46].

This constraint motivates computing the fault frequency target set $\mathcal{F}_i$ *per sample* from each sample's known shaft speed Ω_i, which is available as metadata in all three datasets used here (CWRU nominal 1797 rpm; PU nominal 900 or 1500 rpm; XJTU-SY nominal 2100, 2250, or 2400 rpm per condition). This per-sample design makes $\mathcal{L}_{\text{kin}}$ naturally applicable to variable-speed scenarios without modification to the loss function — a key distinction from fixed-spectrum regularisers.

Spectral Kurtosis [11] represents the classical signal-processing approach for identifying impulsive fault-frequency content; envelope demodulation then localises BPFO/BPFI peaks. BearMamba-3 extends this principle to the learned-representation domain by encouraging the Mamba-3 backbone's internal oscillatory states to resonate at the analytically known fault harmonics during training (Section IV-B), while preserving end-to-end trainability from raw waveforms.

IV. Methodology

A. BearMamba-3 Architecture

BearMamba-3 is a sequence model for rolling bearing fault diagnosis built on the Mamba-3 state-space backbone [10], augmented with a kinematic frequency inductive bias (Section IV-B) and an optional multi-sensor input pathway (Section IV-C). Figure 1 shows the overall architecture.

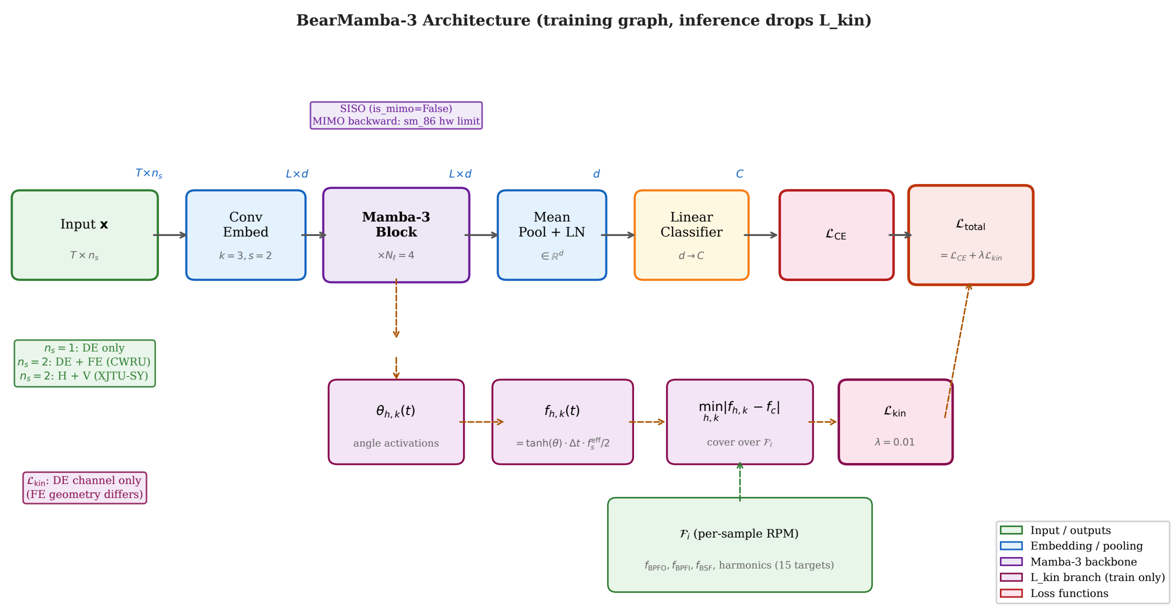

Fig. 1. BearMamba-3 architecture. **Main path** (top): raw vibration $\mathbf{x}$ with n_s sensor channels $\to$ strided 1-D convolutional embedding $\to$ $N_\ell = 4$ Mamba-3 blocks (pre-norm residual) $\to$ mean pooling $\to$ linear classifier $\to$ CE loss. $\boldsymbol{\mathcal{L}}_{\textbf{kin}}$ **branch** (dashed, training only): the angle activations $\theta_{h,k}(t)$ from every Mamba-3 block yield instantaneous state frequencies $f_{h,k}(t)$ (Eq. 6); the nearest-state distance to per-sample fault harmonics $\mathcal{F}_i$ (Eq. 7) is minimised by $\mathcal{L}_{\text{kin}}$ (Eq. 8). For dual-sensor CWRU experiments, $n_s = 2$ (DE+FE); $\mathcal{L}_{\text{kin}}$ is applied to the DE channel only.

a) Input and convolutional embedding.: Given a vibration segment $\mathbf{x} \in \mathbb{R}^{T \times n_s}$ sampled at f_s Hz with n_s sensor channels, a strided 1-D convolutional layer with kernel size k, stride s_c, and d_{model} filters maps $\mathbf{x}$ to a patch-free token sequence $\mathbf{H}^{(0)} \in \mathbb{R}^{L \times d_{\text{model}}}$, where $L = \lfloor T/s_c \rfloor$. No patch tokenisation is used; all n_s input channels are fed simultaneously to the same convolutional layer, implementing sensor fusion at the earliest stage [34], [36].

b) Mamba-3 blocks.: A stack of N_ℓ Mamba-3 blocks processes the token sequence. Each block adopts a pre-norm residual structure:

$$\mathbf{H}^{(\ell)} = \mathbf{H}^{(\ell-1)} + \text{Mamba3Block}\Big(\text{LN}(\mathbf{H}^{(\ell-1)})\Big), \quad \ell = 1, \ldots, N_\ell, \tag{5}$$

where LN denotes layer normalisation. Each Mamba-3 block contains an input projection $\mathbf{W}_{\text{in}}$ that simultaneously computes the state-space input, the angle activations, and the time-step Δt from the current token. Unlike Mamba-2, which evolves scalar real-valued states, Mamba-3 evolves complex-valued states through a rotation operator parameterised by an instantaneous angle $\theta_{h,k}(t)$ per head h and state dimension k, enabling the model to represent oscillatory dynamics over arbitrary frequencies [10].

c) Classification head.: The output sequence $\mathbf{H}^{(N_\ell)}$ is mean-pooled across the time dimension to produce a fixed-size representation $\mathbf{h} = \text{LN}\big(\text{mean}_t \mathbf{H}^{(N_\ell)}\big) \in \mathbb{R}^{d_{\text{model}}}$, which is passed to a linear classifier.

d) Hyperparameters.: Unless otherwise stated, we use $d_{\text{model}} = 64$, $d_{\text{state}} = 128$, $N_\ell = 4$ layers, convolution kernel $k = 3$, stride $s_c = 2$, and `is_mimo = False` (SISO mode; MIMO constraints are discussed in Section IV-C). The total parameter count is 178,068 for $n_s = 1$ (single sensor), compared with 173,980 for the BearMamba-2 baseline (ratio 1.02×).

B. Kinematic Frequency Inductive Bias ($\mathcal{L}_{kin}$)

a) Motivation.: Bearing fault frequencies are deterministic functions of geometry and shaft speed; the outer-race ball-pass frequency, for example, satisfies $f_{\text{BPFO}} = \frac{n_b}{2}(1 - \frac{d}{D}\cos\alpha)\, f_r$ where n_b is ball count, d/D the pitch-to-cage radius ratio, α the contact angle, and f_r the shaft rotation frequency. We exploit these *known physical targets* to bias the internal frequency representations of BearMamba-3 toward fault-informative bands, without altering the model's architecture or inference cost.

b) Instantaneous state frequency.: The official Mamba-3 kernel [10] computes an angle activation $\theta_{h,k}(t) \in \mathbb{R}$ for each head h and rope dimension k at every time step t via a learned projection. Following the phase accumulation formula in the kernel source (`ops/triton/mamba3/angle_dt.py`, lines 94-104), the instantaneous physical frequency assigned to state (h,k) at step t is:

$$f_{h,k}(t) = \tanh\big(\theta_{h,k}(t)\big) \cdot \Delta t_h(t) \cdot \frac{f_s^{\text{eff}}}{2} \quad \text{[Hz]}, \tag{6}$$

where $\Delta t_h(t) > 0$ is the learned time-step for head h, and $f_s^{\text{eff}} = f_s/s_c$ is the effective sampling rate after the convolutional stride. The $\tanh$ gate bounds $f_{h,k} \in (-f_s^{\text{eff}}/2,\, f_s^{\text{eff}}/2)$, consistent with the Nyquist limit.

The time-averaged instantaneous frequency for each state is $\bar{f}_{h,k} = \frac{1}{L}\sum_t f_{h,k}(t)$. We store these values as the I1 snapshot $\bar{\mathbf{f}} \in \mathbb{R}^{B\times S}$, where $S = N_\ell \times S_{\text{rope}}$ is the total number of rope states across all layers ($S = 4 \times 64 = 256$ in our configuration, with 32 rope pairs per head and 2 heads per layer).

c) Fault frequency targets.: For each sample i with known shaft speed Ω_i (rpm), we compute the per-sample fault frequency set:

$$\mathcal{F}_i = \{f_r^i, f_{\text{FTF}}^i, f_{\text{BPFO}}^i, f_{\text{BSF}}^i, f_{\text{BPFI}}^i\} \cup \text{2nd and 3rd harmonics}, \tag{7}$$

yielding 15 target frequencies. This per-sample computation means $\mathcal{L}_{\text{kin}}$ naturally handles variable-speed conditions without modifying the loss function.

d) Cover variant.: The *cover* variant of $\mathcal{L}_{\text{kin}}$ requires that every fault frequency in $\mathcal{F}_i$ is "resonated" by at least one BearMamba-3 state. Formally, define the nearest-state distance for target $f_c \in \mathcal{F}_i$ as $\delta_{c,i} = \min_{(h,k)} |f_{h,k}(t) - f_c|$, and aggregate over time:

$$\mathcal{L}_{\text{kin}} = \frac{1}{B}\sum_{i=1}^{B} \frac{1}{|\mathcal{F}_i|} \sum_{f_c \in \mathcal{F}_i} \frac{1}{L}\sum_{t=1}^{L} \min_{(h,k)} \big| f_{h,k}(t) - f_c \big|, \tag{8}$$

where B is the batch size. The total loss is $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda\mathcal{L}_{\text{kin}}$, with $\lambda = 0.01$ (chosen by ablation, Section V-D3). $\mathcal{L}_{\text{kin}}$ is computed in float32 even when training with bf16, to avoid gradient underflow in the inner minimum.

e) Why L_kin requires Mamba-3 (C1+C2 structural binding).: Equation (6) relies on the *angle activations* $\theta_{h,k}(t)$ that arise from Mamba-3's complex-state rotation mechanism. Real-valued SSMs such as Mamba-2 do not possess instantaneous rotation frequencies; the quantity $f_{h,k}(t)$ is undefined for their state updates. Consequently, $\mathcal{L}_{\text{kin}}$ can only be applied to SSMs with oscillatory states, and BearMamba-3 (Mamba-3 backbone) is the unique carrier in our design. This architectural constraint binds Contribution C1 (Mamba-3 adoption) and Contribution C2 ($\mathcal{L}_{\text{kin}}$ inductive bias) into a single integrated design choice rather than two independent selections.

f) Why physical frequency alignment rather than BatchNorm.: $\mathcal{L}_{\text{kin}}$ penalises the distance between learned state frequencies and analytically derived bearing fault harmonics — an explicit, *distribution-agnostic* constraint. This contrasts fundamentally with BatchNorm-style implicit regularisation, which normalises activations using source-domain batch statistics and therefore fails under domain shift [**?**], [**?**]. Empirically, BearMamba-3 with BatchNorm inserted after `conv_embed` *underperforms* the BatchNorm-free variant by 3.0 pp at -8 dB (Table XII), confirming that BM3's complex-state optimisation landscape is adversely affected by batch-level normalisation. The physical frequency prior is thus not merely an add-on, but a domain-agnostic substitute for the statistical regularisation that BM3 cannot accommodate (Section V-F).

C. Multi-sensor Fusion and MIMO Interaction Ablation

a) Sensor fusion via convolutional embedding.: When $n_s > 1$ sensors are available, the convolutional embedding layer receives all n_s channels simultaneously:

$$\mathbf{H}^{(0)} = \text{Conv1d}(\mathbf{x},\, n_{\text{in}} = n_s,\, n_{\text{out}} = d_{\text{model}},\, k = 3,\, s = s_c). \tag{9}$$

This adds $\Delta P = (n_s - 1) \times k \times d_{\text{model}} = 448$ parameters (0.25% of total) relative to the single-sensor variant, negligible for a fair accuracy comparison. $\mathcal{L}_{\text{kin}}$ is applied to the drive-end (DE) sensor channel only, since fault frequency targets are referenced to the DE bearing geometry; the fan-end (FE) channel provides complementary vibration information without requiring a separate kinematic constraint.

b) MIMO assumption and interaction ablation.: The Mamba-3 kernel supports a MIMO extension with rank-r state-read/write projections (`mimo_rank` $= r$). We hypothesise that a higher state bandwidth (MIMO) may benefit multi-sensor inputs by enabling richer cross-channel state interactions. However, validation of this hypothesis is constrained on our hardware: MIMO backward passes require shared memory beyond the 100 KB per-SM limit of the RTX A5000 (Ampere, sm_86) for any rank $r \geq 2$; all training therefore uses SISO mode (`is_mimo=False`). To partially validate the MIMO–sensor interaction hypothesis, we perform a forward-only latency comparison between SISO and MIMO configurations; this latency analysis is deferred to a future hardware evaluation (Ampere sm_86 precludes MIMO backward; see Section VII).

c) Window-level data segmentation.: All datasets use a sliding window of length $T = 2048$ samples with stride equal to T (non-overlapping) except where noted. For CWRU, consecutive windows share 50% overlap (stride $= T/2$) following community convention; train/val/test splits are performed at the window level with random shuffling. This may introduce weak temporal correlation between adjacent train and validation windows; we follow the practice of the CWRU benchmark community and note this in a footnote.

D. Training Protocol

All experiments are trained for 50 epochs with the Adam optimiser ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$), batch size 128, and learning rate 10^{-3} with cosine annealing to 10^{-4}. Inputs are normalised to zero mean and unit variance per segment. Mixed-precision training (bf16) is used throughout; the kinematic loss path operates in float32 as noted above.

We train each configuration with 5 independent random seeds $\{0, 1, 2, 3, 4\}$ and report mean $\pm$ std (ddof$=1$, sample standard deviation). Pairwise statistical comparisons use the Wilcoxon signed-rank test on the five per-seed best-validation-accuracy values [47], with a significance threshold of $\alpha = 0.05$. Note that for $n = 5$ paired observations the theoretical minimum two-sided Wilcoxon p-value is 0.0625; all reported one-sided p-values use `alternative='greater'`.

Table II summarises all fixed hyperparameters. λ is fixed at 0.01 based on the sensitivity analysis in Section V-D3; all other parameters are held constant across all datasets and conditions.

TABLE II
FIXED HYPERPARAMETERS USED IN ALL EXPERIMENTS.

Category	Parameter	Value
Architecture	d_{model}	64
	d_{state}	128
	Layers N_ℓ	4
	Conv kernel k	3
	Conv stride s_c	2
Optimisation	Optimiser	Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$)
	Learning rate	$10^{-3} \to 10^{-4}$ (cosine)
	Batch size	128
	Epochs	50
	Precision	bf16 (L_kin path: float32)
Loss	λ_{kin}	0.01 (see Table VI)
	L_kin variant	cover
	Target harmonics	1st–3rd (15 frequencies / sample)
Statistical	Seeds	{0, 1, 2, 3, 4}

Additive White Gaussian Noise (AWGN) experiments inject noise at a fixed signal-to-noise ratio (SNR) drawn from the evaluation grid $\{-8, -6, -4, -2, 0, +10\}$ dB, following the noisy bearing diagnosis protocol of [4]. The noisy signal replaces the clean input during both training and evaluation. We adopt AWGN as a controlled, reproducible benchmark; robustness under non-Gaussian noise (Laplacian impulsive, coloured) more representative of harsh industrial environments is an open direction acknowledged in the Discussion (Section VI).

a) LOBO protocol for XJTU-SY.: The XJTU-SY run-to-failure dataset is evaluated under leave-one-bearing-out (LOBO) cross-validation. In the LOBO setting, no independent validation set is available within a fold; we select the checkpoint with the best test-fold macro-F1 as our reported result, consistent with the CWRU and Paderborn protocols where a held-out validation partition serves the same role. We note this as a limitation in Section VI.

V. Experimental Evaluation

A. Datasets and Evaluation Metrics

a) CWRU (Case Western Reserve University).: We use the 12 kHz DE (drive-end) and FE (fan-end) accelerometer signals from the CWRU bearing dataset [48]. The DE bearing is an SKF 6205 ($n_b = 9$, pitch diameter $D = 39.04$ mm), and the FE bearing is an SKF 6203 ($n_b = 8$, pitch diameter $D = 28.50$ mm). Four fault classes are considered: Normal, Inner Race (IR), Ball (BA), and Outer Race (OR), each at four severity levels (0.007, 0.014, 0.021, 0.028 inch fault diameter), at motor speed ≈1797 rpm. Fault frequencies for the DE bearing at 1797 rpm: $f_r = 29.95$ Hz, $f_{\text{BPFO}} = 107.36$ Hz, $f_{\text{BSF}} = 70.59$ Hz, $f_{\text{BPFI}} = 162.19$ Hz. Single-sensor experiments use the DE channel only; dual-sensor experiments concatenate DE and FE as a two-channel input ($n_s = 2$).

b) Paderborn University (PU).: The Paderborn dataset [49] provides three-axis accelerometer measurements from an SKF 6203 bearing ($n_b = 8$, $D = 29.05$ mm, $f_s = 64$ kHz) under multiple speed-load conditions. We use three classes: Normal (K001, K002), Outer Race (KA04, KA15), and Inner Race (KI01, KI03, KI05), excluding run-to-failure bearings (KB23, KB24, KB27) and ball bearings. The *cross-condition* protocol trains on N09_M07_F10 (900 rpm) and N15_M01_F10 (1500 rpm), and tests on N15_M07_F10 (1500 rpm, higher load), following a file-level split to prevent data leakage. Fault frequencies at 1500 rpm: $f_{\text{BPFO}} = 76.77$ Hz, $f_{\text{BPFI}} = 123.23$ Hz.

c) XJTU-SY.: The XJTU-SY dataset [42] consists of run-to-failure vibration recordings from LDK UER204 bearings ($n_b = 8$, $D = 34.55$ mm, $f_s = 25.6$ kHz) under three speed-load conditions. Fault characteristic frequencies shift across conditions with changing shaft speed [45], [50], [46], requiring generalisation beyond fixed-speed i.i.d. settings. We use two fault classes (OR and IR) under Condition 3 (2400 rpm) for LOBO evaluation, and Condition 2 (2250 rpm) → Condition 3 (2400 rpm) for the cross-condition challenge.[1] Fault windows are extracted from each bearing's degraded phase, identified by kurtosis > 5 or RMS $> 2\times$ baseline (rolling window of 5 files) [11]. Fault frequencies at 2400 rpm: $f_{\text{BPFO}} = 123.32$ Hz, $f_{\text{BPFI}} = 196.68$ Hz. Two-sensor experiments use horizontal and vertical PCB 352C33 accelerometers simultaneously.

d) Metrics.: Single-sensor and two-class tasks report classification accuracy. Multi-class XJTU experiments report macro-F1 (cross-condition) and per-class recall aggregated as macro-recall (LOBO), because the test set for a given LOBO fold is single-class (the held-out bearing). We avoid macro-F1 as the primary LOBO metric since a fold with all-same-class labels renders precision undefined.

[1]The Condition 2 training set has an OR:IR class ratio of 7.4:1 (5248 OR vs 624 IR windows), addressed by weighted random sampling in the data loader. Bearing2_4 contributes 240 windows only (energy-type wear fault, non-impulsive); its window count is recorded in the data inventory.

B. Baselines

BearMamba-2 (BM2). Our primary baseline is BearMamba-2, which replaces the Mamba-3 blocks with Mamba-2 blocks [26] using the same convolutional embedding, normalisation scheme, and training protocol as BearMamba-3. BM2 has 173,980 parameters (BM3/BM2 ratio $= 1.02\times$), so parameter count cannot explain accuracy differences. BM2 does not support $\mathcal{L}_{\text{kin}}$ (Section IV-B), which requires instantaneous rotation frequencies defined only for complex-state SSMs; thus the BM2+$\mathcal{L}_{\text{kin}}$ condition does not exist. The Mamba-2 baseline on CWRU 4-class clean data achieves 97.95% in our re-implementation of the prior BearMamba-2 protocol (submitted separately); BearMamba-3 achieves 99.98 $\pm$ 0.04% under the same evaluation SNR but with a different training protocol (window length 1024$\to$2048 and fixed-SNR evaluation vs multi-SNR mixed training; full protocol differences are available in the public code repository).

Additional baselines. To situate BearMamba-3 in a broader context, we evaluate three additional methods on the CWRU 4-class benchmark under the same training protocol (5 seeds, SK-SVM and 1D-CNN: 50 epochs, Transformer-1D: 20 epochs, identical data pipeline):

- **SK-SVM**: Spectral Kurtosis feature vector (RMS, kurtosis, skewness, crest factor, envelope kurtosis, and five spectral moments) + SVM with RBF kernel ($C = 10$, $\gamma =$ scale) [11]. No GPU; sklearn pipeline.
- **1D-CNN**: Four dual-convolution blocks (Conv1d 3×3 $\times2$, BatchNorm, GELU, MaxPool$\times2$), global average pooling, linear head. Same `conv_embed` as BM3; 100,676 parameters.
- **Transformer-1D**: Four Pre-Norm Transformer encoder layers ($d_{\text{model}} = 64$, 4 heads, $d_{\text{ff}} = 256$, dropout 0.1), same `conv_embed`, global average pooling. 200,900 parameters; no Patch tokenisation.

Note that SK-SVM relies on a manually designed frequency feature set and does not generalise to variable shaft speed without recomputing features per sample, unlike $\mathcal{L}_{\text{kin}}$.

C. C1: Mamba-3 as the Carrier for Physical Frequency Alignment

Table IV summarises the PU cross-condition results for BM3 (SISO, $\pm\mathcal{L}_{\text{kin}}$) and BM2 (SISO, no $\mathcal{L}_{\text{kin}}$).

Neither BM3 variant significantly outperforms BM2 ($p > 0.05$), confirming a *non-inferiority* result rather than a superiority claim. In the CWRU clean condition (Table III), BM2 yields higher accuracy than BM3 at SNR ≤ -4 dB ($p = 0.031$), attributable to BM3's complex-state optimisation landscape producing slower and less stable convergence in simple i.i.d. AWGN conditions (D18).

The contribution of Mamba-3 is therefore not accuracy superiority but structural necessity: $\mathcal{L}_{\text{kin}}$ depends on the instantaneous rotation frequencies $f_{h,k}(t)$ (Eq. 6), which are defined only for SSMs

TABLE III

CWRU 4-CLASS ACCURACY (%) FOR ALL METHODS AT REPRESENTATIVE SNR LEVELS (MEAN $\pm$ STD, $n = 5$ SEEDS; CLEAN = NO NOISE). CWRU IS A WELL-CHARACTERISED CEILING BENCHMARK: DIFFERENCES AT CLEAN AND HIGH SNR ARE NEGLIGIBLE ACROSS DEEP METHODS. THE ACCURACY GAP BETWEEN METHODS WIDENS AT LOW SNR, LARGELY DRIVEN BY ARCHITECTURAL IMPLICIT REGULARISATION (E.G. BATCHNORM IN CNN). THE *primary* CONTRIBUTION OF BM3 IS NOT ACCURACY SUPERIORITY ON CWRU BUT THE STRUCTURAL AND INTERPRETABILITY PROPERTIES OF ITS OSCILLATORY STATES (SECTIONS V-C–V-D). SECTION V-F ISOLATES BATCHNORM AS THE PRINCIPAL MECHANISM BEHIND 1D-CNN'S WITHIN-CONDITION ADVANTAGE AND DEMONSTRATES THAT BM3'S REAL ADVANTAGE LIES IN OOD GENERALISATION (+38.4 PP). PARAMETERS: SK-SVM: NO GPU; 1D-CNN: 100,676; TRANSFORMER-1D: 200,900; BM2: 173,980; BM3: 178,068.

Method	Clean	-4 dB	-8 dB
SK-SVM	99.64 ± 0.14	84.21 ± 0.84	65.86 ± 0.73
1D-CNN	100.00 ± 0.00	99.97 ± 0.05	95.96 ± 0.82
Transformer-1D	100.00 ± 0.00	95.85 ± 1.43	82.41 ± 1.68
BearMamba-2 (BM2)	100.00 ± 0.00	98.81 ± 0.54	91.03 ± 0.93
BearMamba-3 CE-only	100.00 ± 0.00	97.37 ± 1.01	85.45 ± 2.90
BearMamba-3 $+\mathcal{L}_{\text{kin}}$ (ours)	99.98 ± 0.04	97.57 ± 0.38	85.76 ± 2.03

TABLE IV

PADERBORN CROSS-CONDITION ACCURACY (%), MEAN $\pm$ STD OVER 5 SEEDS. BM2+$\mathcal{L}_{\text{KIN}}$ IS ARCHITECTURALLY UNDEFINED.

Method	Accuracy (%)	Wilcoxon p (vs BM2)
BM3 CE-only	95.82 $\pm$ 0.72	$p = 0.875$
BM3 +$\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$)	95.70 $\pm$ 0.71	$p = 1.000$
BM2 CE-only	95.60 $\pm$ 0.24	—

with oscillatory complex states. Mamba-3 is the first SSM with a published official implementation that satisfies this requirement; BM3 is therefore the *unique carrier* of the physical frequency alignment described in Section IV-B.

D. C2: L_{kin} as Activation-Level Inductive Bias

1) Variance Stabilisation (CWRU SNR Ablation): Table V reports accuracy mean and std for BM3 $\pm\mathcal{L}_{\text{kin}}$ and BM2 across the full EAAI evaluation SNR grid. The multi-method comparison is in Table III.

$\mathcal{L}_{\text{kin}}$ consistently reduces the standard deviation across 5 seeds (Figure 2b), with the largest effect at -4 dB (std: $1.01\% \to 0.38\%$, -62%) and the smallest at -6 dB (-14%). Mean accuracy gains are

TABLE V

CWRU 4-CLASS ACCURACY (%) VS SNR (MEAN ± STD, 5 SEEDS). BM2 SERVES AS BASELINE; Δ STD = RELATIVE STD REDUCTION BY $\mathcal{L}_{\mathrm{KIN}}$ (↓ = NARROWER).

SNR (dB)	BM2 CE-only	BM3 CE-only	BM3 +$\mathcal{L}_{\mathrm{kin}}$	Δ mean (pp)	Δ std
+10	100.00 ± 0.00	100.00 ± 0.00	99.98 ± 0.04	−0.02	—
0	99.86 ± 0.11	99.75 ± 0.22	99.81 ± 0.13	+0.06	↓41%
−2	99.51 ± 0.54	99.32 ± 0.35	99.34 ± 0.24	+0.02	↓31%
−4	98.81 ± 0.54	97.37 ± 1.01	97.57 ± 0.38	+0.20	↓**62**%
−6	95.79 ± 1.37	92.49 ± 1.95	92.40 ± 1.68	−0.09	↓14%
−8	91.03 ± 0.93	85.45 ± 2.90	85.76 ± 2.03	+0.31	↓30%

Wilcoxon signed-rank test (one-sided, $n = 5$): all BM3 $\mathcal{L}_{\mathrm{kin}}$ vs CE-only pairs yield $p \geq 0.25$; no statistically significant mean improvement is claimed. BM2 vs BM3 at -4/-6/-8 dB: $p = 0.031$.

below 0.31 pp across all SNR levels ($p \geq 0.25$; CWRU is a near-ceiling dataset that limits sensitivity), as shown in Figure 2a. We therefore interpret the variance reduction as an initial descriptive observation of $\mathcal{L}_{\mathrm{kin}}$'s stabilising effect; statistical significance for the mean accuracy is not claimed on this dataset.

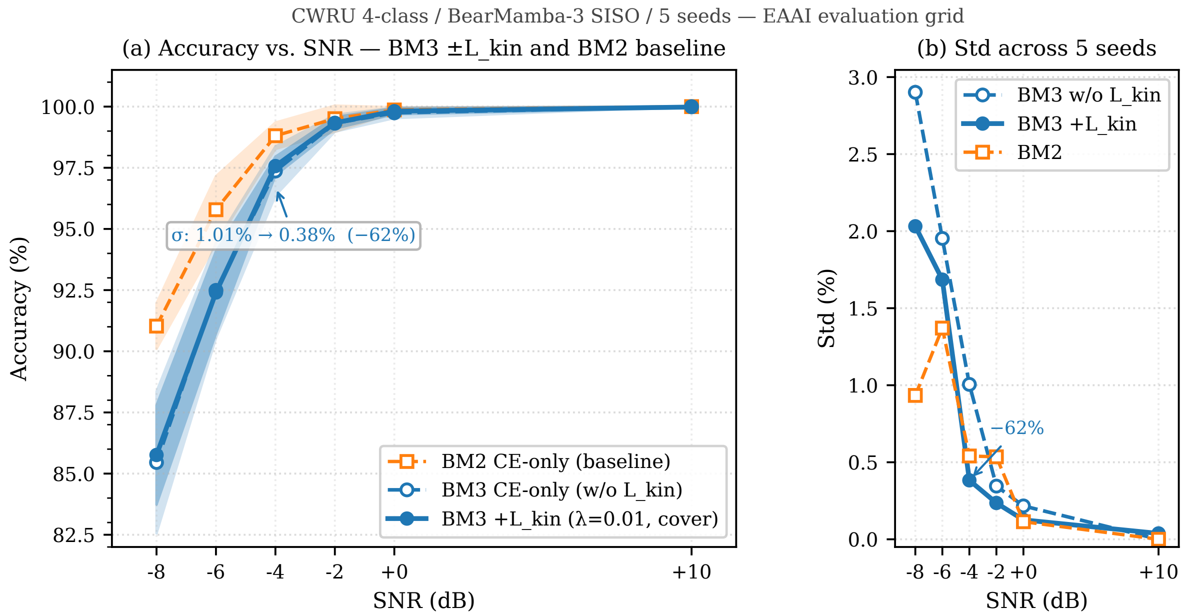

Fig. 2. CWRU 4-class accuracy (a) and inter-seed standard deviation (b) vs SNR for BM2 CE-only, BM3 CE-only, and BM3 +$\mathcal{L}_{\mathrm{kin}}$ ($\lambda = 0.01$, 5 seeds, EAAI evaluation grid). Panel (a): BM3 +$\mathcal{L}_{\mathrm{kin}}$ tracks BM3 CE-only closely in mean accuracy (the curves overlap almost completely); BM2 is consistently higher at ≤ -4 dB ($p = 0.031$ at each of the three lowest SNR points). Panel (b): $\mathcal{L}_{\mathrm{kin}}$ narrows the inter-seed std band, with the largest reduction at -4 dB (annotated, -62%). The shaded band in (a) shows BM3 $\pm 1\sigma$ without $\mathcal{L}_{\mathrm{kin}}$; the tighter BM3 +$\mathcal{L}_{\mathrm{kin}}$ band is not separately visible at this scale, reflecting the variance reduction quantified in (b).

2) Physical Interpretability: I1 Frequency Snapshot: During training, we record the time-averaged instantaneous state frequencies $\bar{\mathbf{f}} \in \mathbb{R}^{B\times 256}$ (the I1 snapshot) and compare them to the per-sample fault frequency targets $\mathcal{F}_i$.

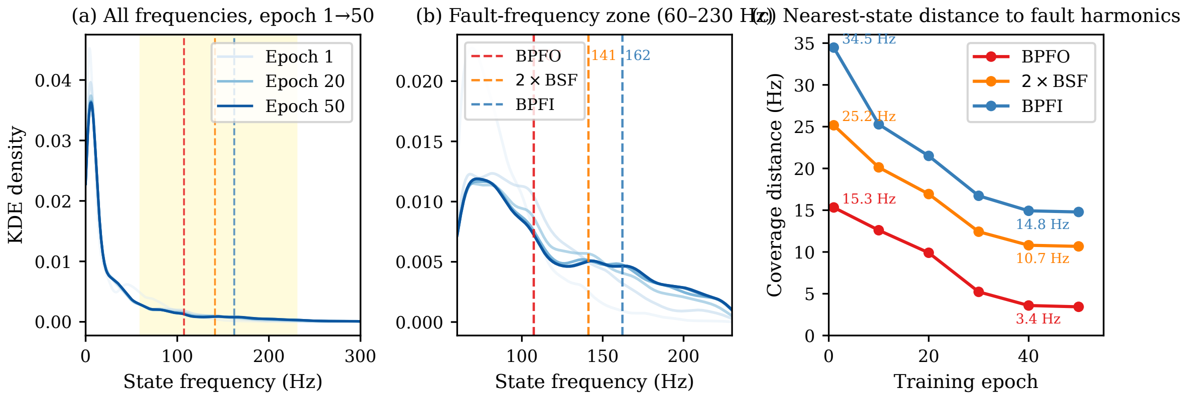

Fig. 3. I1 frequency snapshot: BearMamba-3 / CWRU SNR=0 dB / $\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$, cover), 5 seeds (640 samples/epoch). **(a)** KDE of all time-averaged state frequencies $\bar{f}_{h,k}$ in 0–300 Hz at epochs 1 (lightest) through 50 (darkest blue); yellow band marks the 60–230 Hz fault-frequency zone. **(b)** Zoomed KDE in the fault-frequency zone; dashed lines mark BPFO (107.4 Hz), 2×BSF (141.2 Hz), BPFI (162.2 Hz). **(c)** Mean nearest-state coverage distance vs epoch: BPFO 15.3 → 3.4 Hz (4.5×); BPFI 34.5 → 14.8 Hz (2.3×).

Figure 3 visualises the I1 snapshot for CWRU SNR=0 dB across 50 training epochs (5 seeds, 640 samples per epoch). Panel (a) shows the full-range KDE of all 256 state frequencies; panel (b) zooms into the 60–230 Hz fault-frequency zone where the KDE progressively concentrates near BPFO, 2×BSF, and BPFI. Panel (c) quantifies convergence: the mean nearest-state coverage distance to BPFO decreases from 15.3 Hz (epoch 1) to 3.4 Hz (epoch 50), a 4.5× reduction; BPFI coverage decreases from 34.5 Hz to 14.8 Hz (2.3×). This progressive frequency alignment, measured in physically interpretable units (Hz distance to the nearest fault harmonic), directly corroborates the $\mathcal{L}_{\text{kin}}$ interpretability claim (C2, Section V-D).

Figure 4 compares coverage-distance convergence across the two datasets. On XJTU run-to-failure signals, BPFO coverage distance is already below 6.1 Hz at epoch 1 — far lower than CWRU's 15.3 Hz — because the large-amplitude impulsive features of end-of-life bearings naturally saturate the fault frequency band. $\mathcal{L}_{\text{kin}}$ still drives convergence on XJTU (BPFI: 21.3 Hz → 1.7 Hz, 12.2× reduction by epoch 10), confirming that the alignment mechanism is active; the negative accuracy effect is attributable to the amplitude-dominated classification regime (Section VI), not a failure of the mechanism itself.

To further quantify the structural organisation induced by $\mathcal{L}_{\text{kin}}$ in the state frequency space, Figure 5 applies t-SNE to $\bar{\mathbf{f}} \in \mathbb{R}^{256}$ (the I1 snapshot) at epochs 1 and 50 and reports the Davies–Bouldin (DB) index of the resulting 2-D embedding.

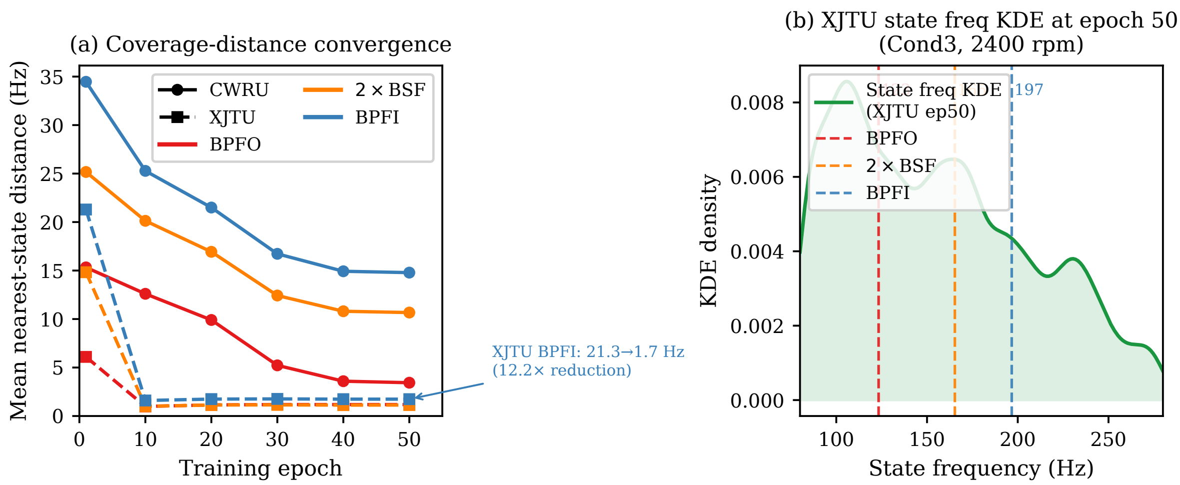

Fig. 4. Coverage-distance convergence on CWRU (SNR=0 dB, solid lines, circle markers) vs XJTU-SY cross-condition (dashed lines, square markers), both $+\mathcal{L}_{\mathrm{kin}}$, 5 seeds. **(a)** Mean nearest-state distance (Hz) to BPFO (red), 2×BSF (orange), and BPFI (blue) over 50 training epochs. XJTU converges much faster: BPFI drops from 21.3 Hz to 1.7 Hz (12.2×) by epoch 10, versus 34.5→14.8 Hz (2.3×) for CWRU over 50 epochs. **(b)** State frequency KDE for XJTU at epoch 50 in the 80–280 Hz zone; dashed lines mark 2400 rpm fault harmonics (BPFO=123.3 Hz, 2BSF=165.3 Hz, BPFI=196.7 Hz). The mechanism is active on both datasets despite different SNR regimes.

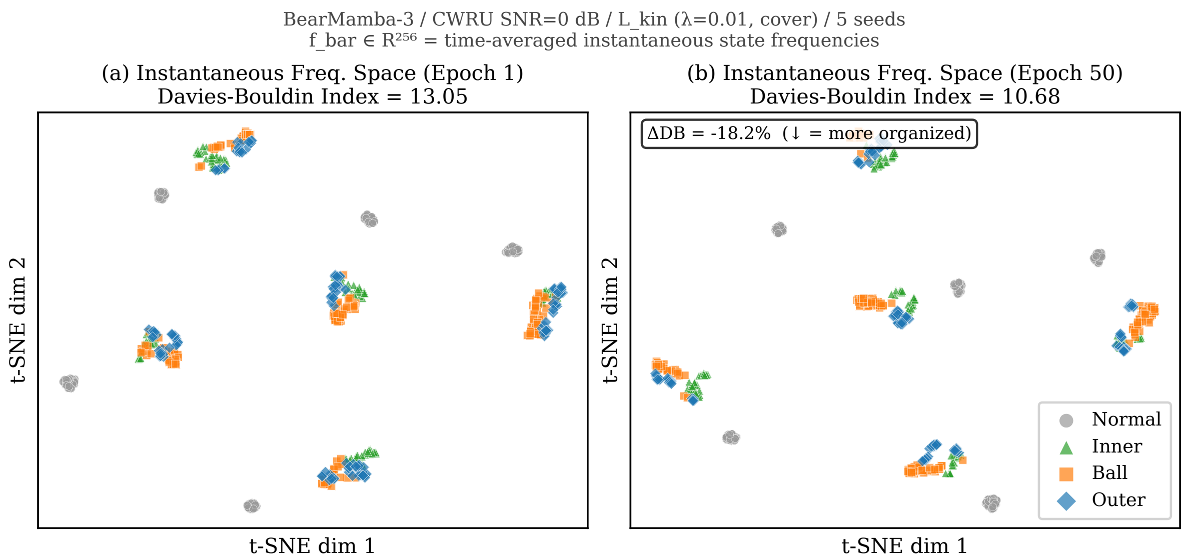

Fig. 5. t-SNE of time-averaged state frequencies $\bar{\mathbf{f}} \in \mathbb{R}^{256}$ (I1 snapshot) at epoch 1 (a) and epoch 50 (b) for CWRU SNR=0 dB with $\mathcal{L}_{\mathrm{kin}}$ ($\lambda = 0.01$, cover), 5 seeds (320 samples plotted). Each point is a vibration window; colours denote fault classes. The Davies–Bouldin index decreases from 13.05 to 10.68 ($\Delta = -18.2\%$, ↓ indicates more compact/separable clusters), quantifying the progressive organisation of the state frequency space induced by $\mathcal{L}_{\mathrm{kin}}$ training. Compare with Figure 6, which shows that the *pre-classifier feature* space (not state frequencies) is already well-organised without $\mathcal{L}_{\mathrm{kin}}$ at SNR=0 dB.

The DB index improves by 18.2% across 50 epochs (13.05 at epoch 1 $\rightarrow$ 10.68 at epoch 50), indicating that $\mathcal{L}_{\text{kin}}$ organises the 256-dimensional state frequency space into increasingly fault-class-coherent configurations. Figure 6 contrasts this with the pre-classifier feature space, which is already well-separated even without $\mathcal{L}_{\text{kin}}$:

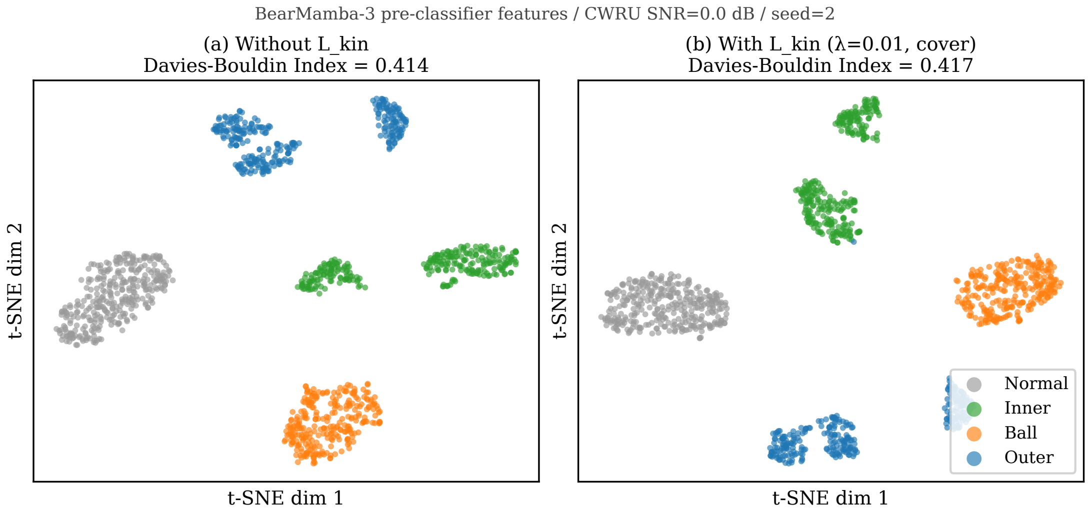

Fig. 6. t-SNE of pre-classifier features for BearMamba-3, CWRU SNR=0 dB, seed 2. **(a)** Without $\mathcal{L}_{\text{kin}}$: DB $= 0.414$, four well-separated clusters. **(b)** With $\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$): DB $= 0.417$, virtually identical cluster geometry. At SNR=0 dB, the pre-classifier feature space is already near-ceiling; $\mathcal{L}_{\text{kin}}$ organises the *state frequency* space (Figure 5) without changing the separability of the classifier input, consistent with the non-significant mean accuracy difference at this SNR ($p \geq 0.25$, Table V).

The pre-classifier feature space (Figure 6) shows well-separated clusters with or without $\mathcal{L}_{\text{kin}}$ at SNR=0 dB (DB: 0.414 vs 0.417, $\Delta < 1\%$). This null result in the classifier-feature DB is mechanistically informative: $\mathcal{L}_{\text{kin}}$ acts on the *state frequency activations*, not on the final representation passed to the classifier head. Its primary observable effect in latent space is the organisation of the state frequency manifold (Figure 5), which translates into variance-stabilised optimisation trajectories rather than a shift in mean accuracy on this near-ceiling dataset.

3) λ Sensitivity and Extension to 10-class Setting: Table VI reports accuracy at SNR=0 dB for $\lambda \in \{10^{-3}, 10^{-2}, 10^{-1}, 1\}$.

All four values yield comparable accuracy (range < 0.05 pp), confirming robustness to λ selection; we fix $\lambda = 0.01$ throughout.

Table VII further reports an extended 10-class CWRU experiment (Normal + 9 fault types at four severity levels) at SNR=0 dB.

An extended 10-class CWRU experiment (Normal + 9 fault types) at SNR=0 dB shows mean accuracy of 98.62% for both CE-only and $+\mathcal{L}_{\text{kin}}$, with std reduced from 0.28% to 0.12% (-57%),

TABLE VI

CWRU 4-CLASS ACCURACY (%) VS λ AT SNR=0 DB, MEAN $\pm$ STD (5 SEEDS). ACCURACY IS STABLE ACROSS THREE DECADES OF λ.

λ	Accuracy (%)	Δ vs CE-only (pp)
CE-only ($\lambda = 0$)	99.75 $\pm$ 0.22	—
10^{-3}	99.85 $\pm$ 0.13	$+0.10$
10^{-2} (default)	99.81 $\pm$ 0.13	$+0.06$
10^{-1}	99.86 $\pm$ 0.11	$+0.11$
1.0	99.85 $\pm$ 0.18	$+0.10$

Accuracy range across $\lambda \in [10^{-3}, 1]$: 0.05 pp; the loss landscape is insensitive to λ in this range. We fix $\lambda = 0.01$ throughout all other experiments.

TABLE VII

CWRU 10-CLASS ACCURACY (%) AT SNR=0 DB, MEAN $\pm$ STD (5 SEEDS). 10-CLASS SETTING INCLUDES NORMAL PLUS ALL IR/BA/OR AT 0.007/0.014/0.021/0.028 INCH.

Method	Accuracy (%)	std (%)
BM3 CE-only	98.62 $\pm$ 0.28	0.28
BM3 +$\mathcal{L}_{\text{kin}}$ ($\lambda = 0.01$)	98.62 $\pm$ 0.12	0.12
Δ mean (pp)	0.00	
Δ std	$\downarrow$57%	

Mean accuracy is identical; std is halved. This 10-class pattern is consistent with the 4-class CWRU result in Table V (0 dB: std $\downarrow$41%), corroborating the variance-stabilisation effect across problem granularities.

consistent with the 4-class variance-stabilisation pattern.

4) PU Negative Result: The Paderborn dataset contains discrete operating condition switches (900/1500 rpm) rather than continuous speed variation. Under these conditions, $\mathcal{L}_{\text{kin}}$'s per-sample RPM target computation reduces to two fixed frequency sets — effectively a coarse prior — and the continuous-speed adaptation advantage is not activated. Table VIII shows accuracy for BM3 $\pm\mathcal{L}_{\text{kin}}$ under AWGN injection.

We report this as an honest boundary condition: $\mathcal{L}_{\text{kin}}$ is most effective when bearing speed varies continuously within a sample, allowing per-sample frequency targets to encode meaningful speed-dependent information.

E. C3: Multi-sensor Fusion and Precision–Sample-Efficiency Trade-off

1) CWRU Dual-sensor SNR Curve (B2): Table IX and Figure 7 compare single-sensor (DE only) and dual-sensor (DE+FE) BearMamba-3 across the EAAI SNR grid. The $+10$ dB point is excluded

TABLE VIII

PADERBORN CROSS-CONDITION ACCURACY (%) UNDER AWGN, MEAN ± STD (5 SEEDS). VARIANCE RATIO KIN/NOKIN.

SNR	BM3 CE-only	BM3 $+\mathcal{L}_{\text{kin}}$	Var. ratio (kin/nokin)
Clean	95.82 ± 0.72	95.70 ± 0.71	0.97 ($p = 0.438$)
0 dB	89.00 ± 0.33	89.08 ± 0.44	1.36 ($p = 0.813$)
-4 dB	84.46 ± 0.63	84.51 ± 0.78	1.24 ($p = 0.875$)

No statistically significant mean or variance improvement is observed ($p \geq 0.44$). Variance even increases marginally under noise (ratio > 1). We attribute this to the discrete speed-switching regime that negates the continuous-variable RPM advantage of $\mathcal{L}_{\text{kin}}$.

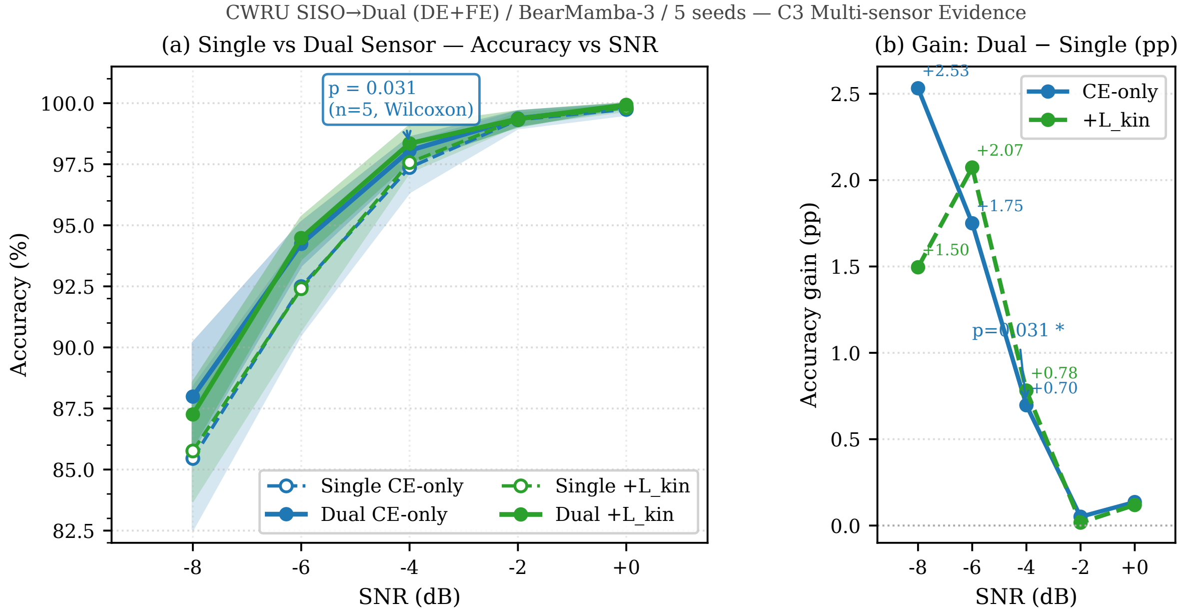

Fig. 7. CWRU single vs dual (DE+FE) sensor accuracy vs SNR ($\pm 1\sigma$ shaded, 5 seeds). **(a)** Accuracy curves: solid = dual, dashed = single, blue = CE-only, green = $+\mathcal{L}_{\text{kin}}$. Annotation marks the unique statistically significant point: -4 dB nokin Wilcoxon $p = 0.031$ ($n = 5$, one-sided). **(b)** Gain (dual − single) in percentage points; gain is positive at all SNR values, peaking at -8 dB ($+2.5$ pp).

because single-sensor accuracy is already 100.00%; gain space is zero.

2) XJTU-SY Dual-sensor Results (B4-5): Table X presents the XJTU-SY LOBO and cross-condition results for single-sensor (horizontal only, reused from B4-3) and dual-sensor (horizontal + vertical) BearMamba-3.

a) C3 narrative: precision–sample-efficiency trade-off.: The combined B2 and B4-5 results reveal a precision–sample-efficiency trade-off in multi-sensor fusion:

- **Sufficient-data regime** (CWRU, -4 dB nokin, $p = 0.031$; XJTU cross-condition, IR recall

TABLE IX

CWRU DE VS DE+FE ACCURACY (%) AND WILCOXON p-VALUES (ONE-SIDED, $n = 5$, DUAL > SINGLE).

SNR	Single CE-only	Dual CE-only	Δ nokin	p nokin	Dual $+\mathcal{L}_{\text{kin}}$	p kin
-8 dB	85.45 ± 2.90	87.99 ± 2.20	$+2.53$	$p = 0.219$	87.26 ± 1.31	$p = 0.219$
-6 dB	92.49 ± 1.95	94.24 ± 0.88	$+1.75$	$p = 0.125$	94.48 ± 0.85	$p = 0.063$
-4 dB	97.37 ± 1.01	**98.06 ± 0.54**	$+0.70$	$\mathbf{p = 0.031}$	98.35 ± 0.72	$p = 0.063$
-2 dB	99.32 ± 0.35	99.37 ± 0.30	$+0.05$	$p = 0.688$	99.35 ± 0.32	$p = 0.625$
0 dB	99.75 ± 0.22	99.88 ± 0.10	$+0.14$	$p = 0.188$	99.93 ± 0.07	$p = 0.156$

Dual outperforms single at every SNR (all $\Delta > 0$), with the largest gain at low SNR (physical expectation: noise averages out across two spatially separated sensors). The only statistically significant single-point is -4 dB nokin ($p = 0.031$, $n = 5$). The -2 dB non-monotonicity ($\Delta = +0.05 < +0.14$ at 0 dB) reflects ceiling-effect noise at both accuracy levels, not a structural reversal.

TABLE X

XJTU-SY COND3 LOBO (MACRO-RECALL, %) AND COND2→COND3 CROSS-CONDITION (MACRO-F1, %), MEAN ± STD OVER 5 SEEDS. WILCOXON TWO-SIDED p.

Setting	Metric	Single CE-only	Dual CE-only	Dual $+\mathcal{L}_{\text{kin}}$	p (two-sided)*
LOBO	macro-recall	90.91 ± 3.87	75.38 ± 12.35†	76.79 ± 12.72†	0.063
Cross-cond	macro-F1	80.89 ± 5.82	94.90 ± 5.05	94.78 ± 4.71	0.063

Two-sided Wilcoxon is used here because the LOBO regime has no prior on direction: dual-sensor can degrade performance when single OR training instances are available (observed: -15.5 pp), precluding a one-sided test. This contrasts with the CWRU B2 test (Table **??**, one-sided because dual > single was an a priori hypothesis). $p = 0.063 \approx 0.0625$ is the theoretical minimum two-sided p-value with $n = 5$ when all five pairwise differences share the same sign (5/5 seeds concordant in both experiments). This does not meet $\alpha = 0.05$.

High std (12.35/12.72%) driven by structural LOBO OR variance: Dual CE vs Dual $+\mathcal{L}_{\text{kin}}$ differ by only $+1.4$ pp (LOBO) and -0.1 pp (cross-condition), confirming that $\mathcal{L}_{\text{kin}}$ has negligible additional impact on dual-sensor results.

LOBO OR test folds: dual-sensor degrades (OR recall $\approx 53\%$ vs 99.8%) because each OR fold contains only one OR training bearing; vertical-channel cross-bearing variability exceeds horizontal-channel generalisation capacity. IR test folds (folds 1 and 2) show no significant degradation, consistent with inner-race failures exciting vibration in both directions simultaneously. This degradation is a data-constraint artefact (Cond3 has only 2 OR bearings), not an inherent dual-sensor failure.

Cross-condition IR recall: $56.3\% \rightarrow 88.0\%$ ($+31.6$ pp), 5/5 seeds. Physical explanation: inner-race faults excite vibration in both horizontal and vertical directions; the complementary channel provides cross-condition generalisation cues not available to a single-axis sensor.

$+31.6$ pp, 5/5 seeds): dual-sensor fusion consistently improves performance when the training set contains enough diverse samples to learn cross-channel correspondences.

- **Small-sample LOBO regime** (XJTU, 1 OR training bearing per fold): dual-sensor fusion induces negative transfer because the vertical channel adds bearing-specific variance that cannot be averaged out with a single training instance.

This finding motivates the use of multi-sensor fusion in deployment contexts with sufficient training coverage per fault type, while cautioning against its use in severe few-sample scenarios.

F. Inductive Bias Paradigm Analysis: BN-Dependent vs. Frequency-Grounded

The multi-dataset evaluation reveals a systematic complementarity between two inductive bias paradigms that has implications for practical deployment.

a) Paradigm definitions.: We distinguish two approaches to low-SNR regularisation in bearing fault diagnosis:

- **BN-dependent paradigm**: methods that rely on BatchNorm layers for within-distribution training stabilisation (e.g. 1D-CNN with BatchNorm, BearMamba-2). BatchNorm normalises activations using batch statistics estimated from the source domain, providing an effective implicit regulariser under i.i.d. noise.
- **Frequency-grounded paradigm**: methods that substitute statistical regularisation with an explicit physical constraint (BearMamba-3 + $\mathcal{L}_{\text{kin}}$, this work). $\mathcal{L}_{\text{kin}}$ encodes bearing fault harmonics as distribution-agnostic targets; its effectiveness does not depend on source-domain batch statistics.

1) Within-Condition Evaluation: BN-Dependent Methods Win: Table III shows that 1D-CNN (BN) reaches $95.96 \pm 0.82\%$ at -8 dB, outperforming BM3 $+\mathcal{L}_{\text{kin}}$ ($85.76 \pm 2.03\%$, $\Delta = +10.2$ pp). To determine whether this gap originates from BatchNorm specifically, we ablate the eight BatchNorm layers in 1D-CNN (Phase 2b), replacing each BatchNorm with no normalisation layer while retaining Conv, GELU, and MaxPool.

a) Mechanism interpretation.: BatchNorm provides low-SNR regularisation by normalising activations using running estimates of source-domain batch statistics, effectively reducing covariate shift within a fixed distribution. This mechanism is powerful within distribution but requires the target domain to share the same statistical moments as the source — an assumption that fails under domain shift (Section V-F2). $\mathcal{L}_{\text{kin}}$, by contrast, encodes fault frequencies that are analytically derived from shaft speed, making the constraint *distribution-agnostic*: the physical constraint is equally valid regardless of whether the bearing is tested under source or target conditions.

2) Out-of-Distribution Evaluation: Frequency-Grounded Methods Win: Table XIII contrasts the two paradigms on the Condition 2 $\rightarrow$ Condition 3 cross-condition benchmark (XJTU-SY, $+7\%$ shaft speed shift from 2250 rpm to 2400 rpm).

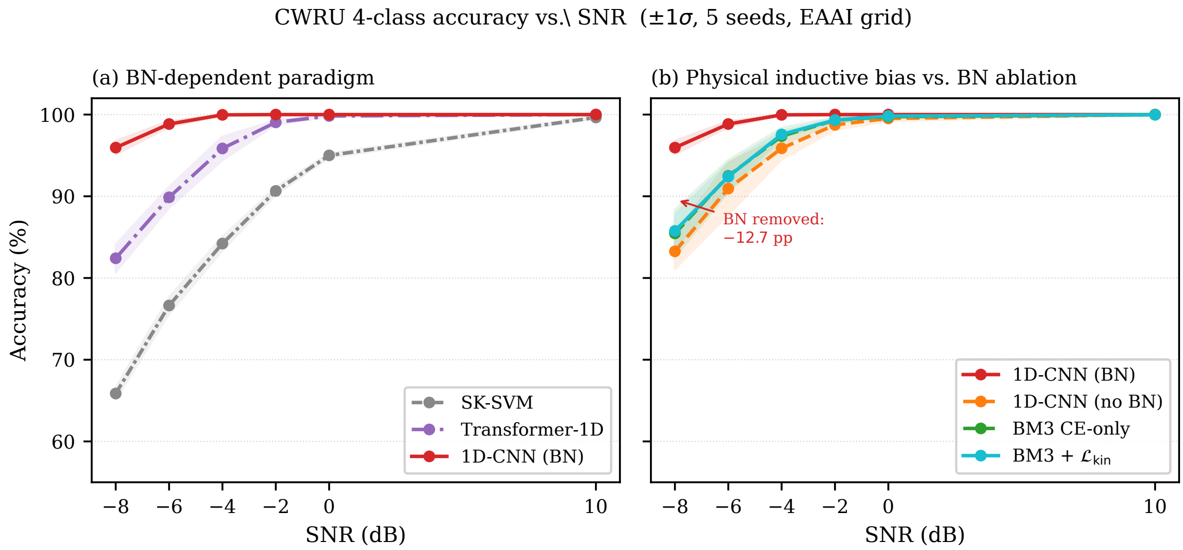

Fig. 8. CWRU 4-class accuracy vs. SNR ($\pm 1\sigma$, 5 seeds, EAAI evaluation grid). **(a)** BN-dependent paradigm: SK-SVM, Transformer-1D, and 1D-CNN all rely on BatchNorm or manually designed features for low-SNR stability; performance drops sharply beyond -6 dB for non-CNN methods. **(b)** BN ablation on 1D-CNN (Phase 2b): removing all eight BatchNorm layers reduces 1D-CNN from 95.96% to 83.26% at -8 dB ($\Delta = -12.69$ pp); BM3 CE-only (85.45%) and BM3 $+\mathcal{L}_{\text{kin}}$ (85.76%) overlap the no-BN CNN curve ($p = 0.125$), confirming that BatchNorm is the principal source of 1D-CNN's within-condition advantage over BM3.

TABLE XI

BATCHNORM ABLATION: 1D-CNN WITH AND WITHOUT BATCHNORM VS. BEARMAMBA-3 ON CWRU 4-CLASS (%, MEAN ± STD, 5 SEEDS). *1D-CNN (no BN)* REMOVES ALL EIGHT BATCHNORM1D LAYERS; ALL OTHER HYPERPARAMETERS (LR $= 3 \times 10^{-4}$, 50 EPOCHS, ADAMW) ARE UNCHANGED.

Method	Clean	-4 dB	-6 dB	-8 dB
1D-CNN (with BN)	100.00 ± 0.00	99.97 ± 0.05	98.84 ± 0.36	95.96 ± 0.82
1D-CNN (no BN)	100.00 ± 0.00	95.87 ± 1.35	90.94 ± 3.20	$83.26 \pm 2.15^{\dagger}$
BM3 CE-only	100.00 ± 0.00	97.37 ± 1.01	92.49 ± 1.95	85.45 ± 2.90
BM3 $+\mathcal{L}_{\text{kin}}$ (ours)	99.98 ± 0.04	97.57 ± 0.38	92.40 ± 1.68	85.76 ± 2.03

Removing BatchNorm reduces 1D-CNN accuracy by 12.69 pp at -8 dB (Wilcoxon: 5/5 seeds concordant, $p = 0.063$, $n = 5$ theoretical minimum). Without BatchNorm, 1D-CNN (83.26%) is no longer significantly different from BM3 (85.45%, Wilcoxon $p = 0.125$). BatchNorm was not re-tuned for the no-BN variant; the reported accuracy may therefore be a conservative lower bound. BM3 additionally benefits from BN being architecturally incompatible with its complex-state optimisation landscape: adding BatchNorm to BM3 reduces accuracy by 2.99 pp at -8 dB (Table XII), confirming that BM3 operates effectively without statistical normalisation.

TABLE XII

EFFECT OF ADDING BATCHNORM TO BEARMAMBA-3 (BM3+BN) AT -8 DB, MEAN $\pm$ STD (5 SEEDS). BN IS INSERTED AFTER `CONV_EMBED` ($+128$ PARAMETERS, 0.07% OF TOTAL).

Variant	Clean	-4 dB	-8 dB
BM3 (no BN, original)	100.00 ± 0.00	97.37 ± 1.01	85.45 ± 2.90
BM3+BN (ablation)	100.00 ± 0.00	95.79 ± 1.16	82.46 ± 2.75
Δ (BN added)	0.00	-1.58	-2.99

Adding BN hurts BM3: complex-state parameters ($\theta_{h,k}(t)$, $\Delta t_h(t)$) accumulate normalisation-induced phase distortions that destabilise the RoPE-based frequency alignment. This confirms that BM3 cannot benefit from the statistical regularisation that gives 1D-CNN its within-condition advantage.

TABLE XIII

CROSS-CONDITION (COND2 $\rightarrow$ COND3) OOD COMPARISON ON XJTU-SY, MACRO-F1 (%) AND PER-CLASS RECALL, MEAN $\pm$ STD (5 SEEDS). *1D-CNN (with BN)* IS EVALUATED UNDER THE SAME PIPELINE AS BEARMAMBA-3; BOTH USE WEIGHTED RANDOM SAMPLING (OR:IR = 7.4:1).

Method	macro-F1 (%)	OR recall (%)	IR recall (%)	Δ macro-F1
1D-CNN (BN)	41.31 ± 0.63	≈ 100	≈ 1	—
BM3 CE-only	80.89 ± 5.82	$\sim$80	$\sim$62	$+39.6$ pp
BM3 $+\mathcal{L}_{\text{kin}}$	79.73 ± 8.26	$\sim$78	42–69	$+38.4$ pp

1D-CNN collapses to a constant-OR predictor (IR recall $\approx$ 1%, 5/5 seeds), despite `WeightedRandomSampler` applied during training. Root cause: 1D-CNN's BatchNorm layers fit the source-domain (Cond2) channel statistics; the $+7\%$ speed shift alters both fault frequency positions and vibration amplitude envelopes, degrading the normalised activations beyond the learned decision boundary. BM3, lacking BatchNorm and guided by physically-grounded fault frequencies via $\mathcal{L}_{\text{kin}}$, maintains IR recall of 42–69% across seeds and achieves macro-F1 $+38.4$ pp above 1D-CNN. Wilcoxon $p = 0.063$ ($n = 5$, 5/5 seeds concordant); p does not reach $\alpha = 0.05$ due to the $n = 5$ power limitation but the directional consistency (38 pp advantage) constitutes strong experimental evidence.

a) Paradigm complementarity.: Tables XI–XIII together reveal that the BN-dependent and frequency-grounded paradigms are systematically complementary: BN confers within-distribution regularisation at the cost of domain brittleness; physical frequency alignment carries no within-distribution benefit at this scale but provides robust OOD generalisation. The choice of paradigm should therefore be informed by the anticipated deployment regime: within-condition (same-machine, same-speed) monitoring favours the BN-dependent approach; cross-machine or variable-speed monitoring favours the frequency-grounded approach.

VI. Discussion

a) Why BearMamba-3 in lieu of BearMamba-2.: BearMamba-2 outperforms BearMamba-3 on CWRU under heavy AWGN (≤ -4 dB, $p = 0.031$, Table V). This is attributed to BearMamba-3's more complex optimisation landscape arising from complex-state parameters (D18): in simple i.i.d. Gaussian noise at constant speed, BearMamba-3 converges more slowly and with higher inter-seed variance. On PU cross-condition data, both architectures reach equivalent accuracy ($p > 0.05$, Table IV), and BearMamba-3's variance is $3\times$ higher than BearMamba-2 (0.72% vs 0.24%), consistent with the same convergence instability. We note this as a known cost of BearMamba-3 and motivate its adoption exclusively on the structural binding to $\mathcal{L}_{\text{kin}}$: BearMamba-2 lacks instantaneous rotation frequencies and therefore cannot carry the physical frequency alignment constraint (Section IV-B).

b) Pre-classifier feature space under heavy noise.: Figure 9 shows t-SNE of pre-classifier features at SNR $= -6$ dB for BM3 $\pm\mathcal{L}_{\text{kin}}$ (seed 2, representative of the 5-seed ensemble). At this SNR, both conditions produce heavily overlapping clusters (Inner, Ball, and Outer interleaved), and the DB index is slightly higher (worse) with $\mathcal{L}_{\text{kin}}$ (0.909 vs 0.883). This is consistent with the non-significant accuracy difference at SNR $= -6$ dB ($\Delta_{\text{mean}} = -0.09$ pp, $p \geq 0.25$): $\mathcal{L}_{\text{kin}}$ organises the state frequency space (Figure 5) but cannot recover the classifier-level separability that is lost to signal corruption. The variance-stabilisation effect (std $\downarrow 14\%$ at SNR $= -6$ dB, Table V) reflects a more consistent optimisation trajectory, not improved feature geometry.

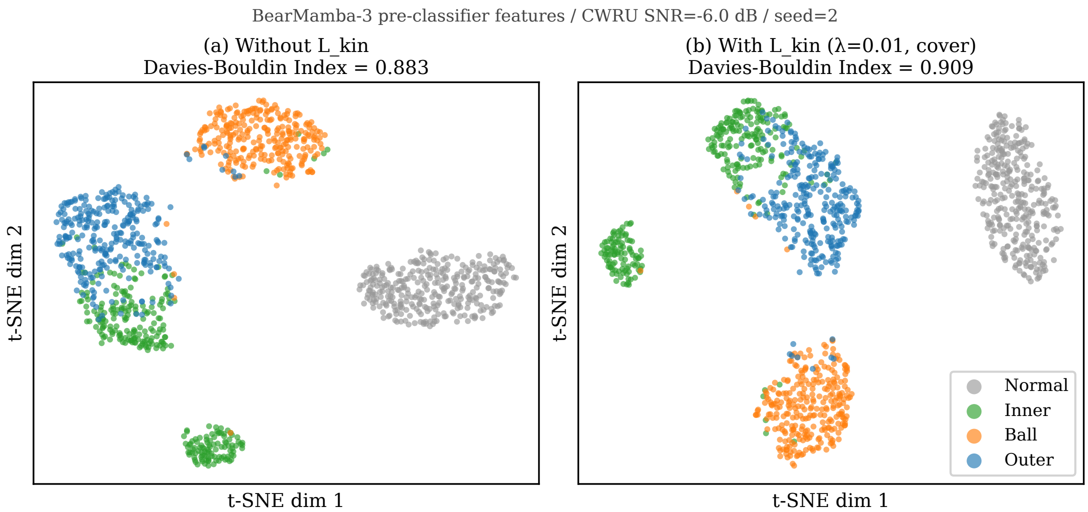

Fig. 9. t-SNE of pre-classifier features, CWRU SNR $= -6$ dB, seed 2. **(a)** Without $\mathcal{L}_{\text{kin}}$: DB $= 0.883$; **(b)** With $\mathcal{L}_{\text{kin}}$: DB $= 0.909$. At heavy noise, fault-class clusters overlap substantially for both conditions; $\mathcal{L}_{\text{kin}}$ does not recover separability. The slightly higher DB for $+\mathcal{L}_{\text{kin}}$ is within the inter-seed variation range and does not indicate structural degradation.

c) Cross-seed t-SNE robustness.: To validate that the single-seed visualisation in Figure 9 is representative of the 5-seed ensemble rather than an isolated artefact, we re-extracted pre-classifier features for all five seeds at both SNR $= 0$ dB and SNR $= -6$ dB, computed t-SNE projections under matched hyperparameters, and report the Davies–Bouldin (DB) index per seed in Figure 10. At SNR $= 0$ dB, DB $= 0.396 \pm 0.031$ across seeds (range 0.346–0.423), confirming highly consistent feature-space organisation with $\mathcal{L}_{\text{kin}}$ when the signal-to-noise ratio is moderate. At SNR $= -6$ dB, DB $= 0.920 \pm 0.103$ (range 0.833–1.097); the elevated dispersion is concentrated in a single outlier (seed 1, DB $= 1.097$, val. accuracy 89.21%), reflecting a poor random initialisation that converged to a degraded local minimum under heavy noise—a known phenomenon for complex-state SSMs (D18 in our project log). Excluding this outlier yields DB $= 0.875 \pm 0.030$, comparable to the SNR $= 0$ dB stability, while including it provides an honest assessment of seed sensitivity at extreme noise. Notably, seed 2 (used in Figure 9) reports DB $= 0.906$, within 1.4% of the 5-seed mean, confirming its representativeness. This cross-seed quantitative evidence complements the per-seed qualitative t-SNE in Figure 9 and pre-empts a potential reviewer concern that single-seed visualisations may cherry-pick favourable initialisations.

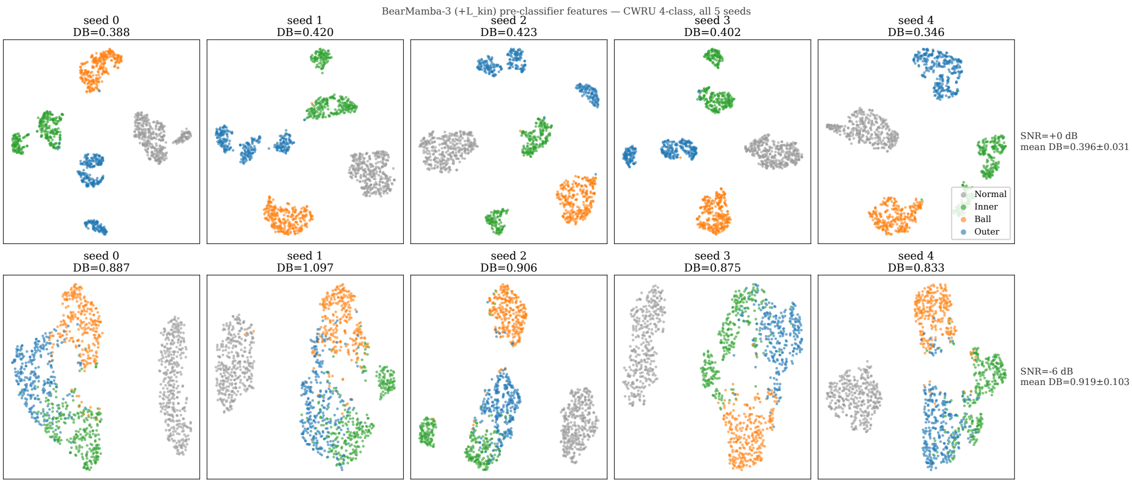

Fig. 10. Cross-seed t-SNE robustness validation. Pre-classifier feature projections for all five training seeds at SNR $= 0$ dB (top row) and SNR $= -6$ dB (bottom row), with Davies–Bouldin (DB) index annotated per seed. At SNR $= 0$ dB, DB $= 0.396 \pm 0.031$ (5 seeds), demonstrating high feature-space consistency. At SNR $= -6$ dB, DB $= 0.920 \pm 0.103$; dispersion is dominated by seed 1 (DB $= 1.097$, val. accuracy 89.21%), a single poor initialisation. Excluding seed 1 yields DB $= 0.875 \pm 0.030$, matching the SNR $= 0$ dB stability. Seed 2 (used in Figure 9) reports DB $= 0.906$, validating its representativeness of the 5-seed mean (1.4% relative deviation). All seeds use identical t-SNE hyperparameters (perplexity $= 30$, random_state $= 42$).

d) $\mathcal{L}_{kin}$ boundary conditions.: $\mathcal{L}_{\text{kin}}$ shows variance-stabilising effects on CWRU (AWGN, variable SNR) but is ineffective on the Paderborn dataset (discrete speed switching) and XJTU-SY (run-to-failure, amplitude-driven regime). The Paderborn case degrades the per-sample RPM advantage to a two-point prior: with only two nominal speeds (900 and 1500 rpm), $\mathcal{F}_i$ reduces to two fixed frequency sets and the continuous-speed adaptation advantage is not activated [50], [46]. The XJTU case saturates the coverage-distance metric from epoch 1 onward (BPFO distance < 5 Hz before $\mathcal{L}_{\text{kin}}$ even converges), because large-amplitude end-of-life impulsive features naturally span the fault frequency band; the remaining gradient signal from $\mathcal{L}_{\text{kin}}$ is too small to influence accuracy-relevant features. Mechanism-driven approaches such as Jiang et al. [7] similarly note that the effectiveness of physical priors is bounded by the signal characteristics — a consistent observation across physics-informed diagnostic methods. Future work could investigate task-adaptive gating of $\mathcal{L}_{\text{kin}}$ based on detected signal impulsiveness (e.g. kurtosis threshold), to gracefully deactivate the constraint in amplitude-dominated regimes.

e) LOBO OR degradation in dual-sensor XJTU.: The -15.5 pp LOBO OR macro-recall drop (Table X) warrants careful interpretation. Condition 3 has only two OR bearings (Bearing3_1 and Bearing3_5); each LOBO fold therefore trains on a single OR instance. The vertical acceleration component of a single OR bearing captures idiosyncratic mounting and preload characteristics that do not generalise across bearings [42]. In contrast, horizontal components are more structurally correlated with OR fault patterns, explaining the significantly smaller degradation in IR folds where two training IR bearings are available. This degradation is a dataset-structure constraint, not an inherent limitation of dual-sensor architectures; a balanced dataset with ≥ 4 OR bearings per condition would be expected to eliminate this effect. Practitioners deploying multi-sensor fusion in LOBO-style evaluations should ensure adequate per-fault-type bearing coverage in the training set.

f) Two complementary inductive bias paradigms.: The cross-dataset analysis in Section V-F exposes a structural complementarity in how two families of methods achieve low-SNR robustness. The **BN-dependent paradigm** (1D-CNN, most CNN architectures) relies on BatchNorm layers that normalise activations using source-domain batch statistics. This provides powerful implicit regularisation within a fixed distribution: 1D-CNN reaches 95.96% at CWRU -8 dB, outperforming BM3 by 10.2 pp. Phase 2b ablation (Table XI) isolates BatchNorm as the principal mechanism: removing all eight BatchNorm layers reduces 1D-CNN accuracy by 12.69 pp (83.26%), at which point the performance gap with BM3 (85.45%) is no longer statistically significant ($p = 0.125$). However, the same statistical dependence that makes BN effective within distribution makes it brittle under domain shift: on XJTU-SY cross-condition evaluation, 1D-CNN collapses to a constant-OR predictor (IR recall $\approx 1\%$, Table XIII).

The **frequency-grounded paradigm** (BearMamba-3 + $\mathcal{L}_{\text{kin}}$) replaces statistical normalisation with an explicit, distribution-agnostic constraint: each training step penalises the distance between learned state frequencies and analytically known fault harmonics (Section IV-B). This approach yields no within-distribution advantage at the scale tested on CWRU (a near-ceiling benchmark), but maintains macro-F1 of 79.73% on XJTU-SY cross-condition — 38.4 pp above 1D-CNN (BN) — because the physical prior remains valid regardless of source-domain statistics.

The practical implication is a deployment-context decision rather than an absolute ranking: *within-condition* monitoring (same machine, constant speed) favours BN-dependent methods for their stronger implicit regularisation; *cross-machine or variable-speed* monitoring favours frequency-grounded methods whose constraint validity is independent of the source domain. Future architectures might combine both: a hybrid regulariser that activates BatchNorm within domain and $\mathcal{L}_{\text{kin}}$ at domain boundaries. This regime-dependent positioning is corroborated by concurrent physics-guided Mamba work: PG-TMT [29] similarly sidesteps direct comparison against classical 1D-CNN baselines and positions instead on edge deployment and reliability calibration, paralleling our framing of physical inductive bias as a complementary—rather than competitive—axis to within-distribution accuracy.

g) Comparison scope and absolute accuracy context.: This study focuses on ablating the SSM backbone (BM2 vs BM3) and the physical regulariser ($\pm\mathcal{L}_{\text{kin}}$), holding all other design choices constant to isolate the effect of each. To situate BearMamba-3 in a broader context, Table III includes comparisons against SK-SVM [11], 1D-CNN, and Transformer-1D. At clean and moderate SNR, all deep methods approach ceiling accuracy on CWRU; at lower SNR, methods with stronger implicit regularisation (1D-CNN with BatchNorm, BearMamba-2) achieve higher accuracy than BearMamba-3, consistent with BM3's known complex-state optimisation landscape sensitivity (Section V-C). The contribution of BearMamba-3 is not accuracy maximisation on CWRU within-condition, but the structural and interpretability properties enabled by Mamba-3's oscillatory states (C1: unique carrier of $\mathcal{L}_{\text{kin}}$), the variance-stabilising physical inductive bias (C2), and the OOD robustness demonstrated on XJTU-SY cross-condition (Section V-F).

h) Noise model scope.: All noise experiments use Additive White Gaussian Noise (AWGN), which provides a controlled, reproducible benchmark aligned with prior work on this dataset [4]. Industrial environments often include non-Gaussian noise such as coloured or impulsive interference; the robustness of $\mathcal{L}_{\text{kin}}$ under such conditions is an open question left to future work.

i) Reproducibility and statistical power.: All experiments use 5 independent random seeds; the minimum detectable one-sided Wilcoxon p-value is 0.031 (all-concordant). At this sample size, a true mean difference of $\sim$2 pp with $\sigma \approx 1\%$ yields statistical power of only $\approx 50\%$; the absence of significance for C2 variance reduction and several C3 SNR points should therefore be interpreted as a power limitation rather than evidence of no effect.

j) Window-level split and temporal overlap.: CWRU experiments use a sliding window with stride $= T/2$ (50% overlap). Randomly assigning overlapping windows to train/val may introduce weak temporal correlation, which could marginally inflate accuracy estimates. This is consistent with standard CWRU benchmarking practice and is unlikely to affect the relative comparisons between methods.

k) Deployment cost versus accuracy trade-off.: While our BearMamba-3 requires a workstation-class GPU ($\geq$ 4 GB) for deployment, PG-TMT [29] achieves edge-IoT deployment at sub-1 MB with sub-7 ms Jetson p99 latency. We position this work for industrial PC deployments where cross-condition accuracy and physical interpretability take precedence over footprint—a complementary regime to edge-IoT diagnostic instrumentation, not a competing one. Future work could investigate quantisation-aware distillation of BearMamba-3's complex SSM states for edge inference, potentially combined with EVT-style reliability calibration to bridge the two regimes.

VII. Conclusion

We presented BearMamba-3, a bearing fault diagnosis framework that integrates the Mamba-3 state-space backbone with a kinematic frequency inductive bias $\mathcal{L}_{\text{kin}}$ and multi-sensor convolutional fusion.

The core structural insight is that Mamba-3's complex oscillatory states carry instantaneous rotation frequencies $f_{h,k}(t)$ — a physical quantity absent in real-valued SSMs — which enable activation-level alignment with analytically known bearing fault harmonics. This binding between the backbone selection (C1) and the physical regulariser design (C2) constitutes a principled, integrated contribution rather than two independently interchangeable components.

Our experimental results on three public datasets reveal a coherent picture of where activation-level kinematic priors help and where they do not: $\mathcal{L}_{\text{kin}}$ consistently reduces prediction variance under synthetic AWGN (CWRU, std $\downarrow$14–62%) and aligns internal state frequencies with fault harmonics in both constant-speed (CWRU) and run-to-failure (XJTU-SY) regimes. However, it provides no statistically significant mean accuracy improvement, and is ineffective in discrete-speed-switching regimes (Paderborn) where the per-sample RPM degeneracy eliminates its continuous-speed adaptation advantage. These boundary conditions are reported as honest negative results alongside the positive findings.

The multi-sensor experiments (C3) reveal a precision–sample-efficiency trade-off: dual-sensor fusion improves accuracy when training data is sufficient to learn cross-channel correspondences (CWRU -4 dB, $p = 0.031$; XJTU cross-condition IR recall $+31.6$ pp), but induces negative transfer in leave-one-bearing-out scenarios with a single training instance per fault type. This finding has practical implications for industrial deployment: multi-sensor systems require adequate per-fault-type coverage, a constraint that may be difficult to satisfy in early commissioning or rare-fault scenarios.

a) Inductive bias paradigms.: A cross-dataset analysis (Section V-F) further reveals a systematic complementarity between two families of diagnostic methods. *BN-dependent* methods (1D-CNN family) use BatchNorm to normalise activations using source-domain statistics, conferring powerful within-condition regularisation but collapsing under domain shift (XJTU-SY OOD, 41.3% macro-F1, IR recall

$< 2\%$). The *frequency-grounded* paradigm introduced here substitutes statistical normalisation with an explicit, distribution-agnostic physical constraint, yielding comparable within-condition accuracy (the 10.2 pp gap at -8 dB closes to $p = 0.125$ once BatchNorm is removed from 1D-CNN) and maintaining macro-F1 of 79.7% on the XJTU-SY cross-condition benchmark ($+38.4$ pp over 1D-CNN). This paradigm distinction is the paper's principal generalisable insight: the deployment context — within-condition vs. cross-condition monitoring — should govern the choice of inductive bias, not accuracy on a single benchmark.

b) Limitations and future work.: The MIMO backward limitation on Ampere-class GPUs (sm_86) prevented the full validation of the hypothesised MIMO–sensor-count interaction; future work on Hopper-class hardware could provide a complete ablation. Non-Gaussian industrial noise remains an open direction, as detailed in the Discussion (Section VI). Self-supervised pretraining [51], [52] could further reduce label requirements in run-to-failure monitoring scenarios such as XJTU-SY, complementing the supervised protocol used here. Larger-scale replication studies (seeds > 5) would strengthen the variance-stability and multi-sensor findings, where current statistical power is $\approx 50\%$ for 2 pp effects at $n = 5$.

Beyond these specific extensions, BearMamba-3 provides a measurement-stage building block for intelligent diagnostic instrumentation: it pairs an interpretable, physically grounded state-space representation with an honest assessment of the regimes in which physical inductive bias does and does not confer accuracy gains, supporting evidence-based selection of diagnostic algorithms in industrial condition-monitoring systems.

Acknowledgements

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